TE 456 - Satellite Communication & Navigation Systems

240 Multiple-Choice Questions from the 24 Presentation Decks

Ten questions per deck. The correct option is marked with a tick, and the reason follows each answer.

Timing advance and frequency offset compensation in LEO NTN

Group 1 deck, slides 2 to 9

1. A user device computes its timing advance from satellite ephemeris and its own GNSS position, and applies it before sending its first preamble. Which stage of the correction chain is this?

- **A. Open-loop timing advance ✓**
- B. Closed-loop frequency offset compensation
- C. The Timing Advance command procedure
- D. Residual tracking by the numerically controlled oscillator

Answer: A. Open-loop timing advance

Open loop means proactive: the device predicts the correction from geometry it already knows, and applies it before it has ever transmitted. The closed loop only starts once the network has measured a real uplink.

2. Why can closed-loop timing advance never remove the timing error completely?

- A. The MAC control element cannot carry a large enough correction value
- **B. The correction is based on geometry that is at least one round trip old ✓**
- C. The base station measures arrival time with limited resolution only
- D. The satellite clock and the device clock use different time references

Answer: B. The correction is based on geometry that is at least one round trip old

Feedback always trails the true geometry by at least one round trip. On a LEO link the geometry moves during that round trip, so a residual error is left behind no matter how accurate the measurement was.

3. In a transparent payload the end-to-end Doppler is roughly double that of the service link alone. What causes the doubling?

- A. The uplink and the downlink carriers are at different frequencies
- B. The signal is sampled twice, once on board and once at the gateway
- **C. The feeder link between the satellite and the gateway adds its own shift ✓**
- D. Inter-satellite links re-transmit the signal to a second spacecraft

Answer: C. The feeder link between the satellite and the gateway adds its own shift

A transparent payload does not demodulate. The signal crosses the service link and then the feeder link, and each moving segment contributes a Doppler shift, so the two add.

4. The device predicts Doppler as $f_D = (V_r / c) \times f_c$. What does V_r stand for?

- A. The full orbital speed of the satellite
- B. The ground speed of the user device
- C. The rate at which the elevation angle is changing
- **D. The component of relative velocity along the line of sight ✓**

Answer: D. The component of relative velocity along the line of sight

Doppler responds only to the closing or opening rate along the line of sight. At the zenith of a pass the satellite is moving fast but almost none of that motion is radial, so the shift crosses zero.

5. Which pair of inputs does the open-loop stage need before the device has transmitted anything at all?

- **A. Satellite ephemeris and the device's own GNSS position ✓**
- B. A Timing Advance command and an AFC error estimate
- C. Received signal strength and the current beam identity
- D. Feeder link Doppler and the gateway clock offset

Answer: A. Satellite ephemeris and the device's own GNSS position

Two positions give a range, and a range gives a delay and a radial velocity. Everything else in the list is either a measurement the network has not made yet or a quantity the device cannot see.

6. A stale ephemeris file degrades performance. Which stage suffers first and worst?

A. The closed loop, because the correction command becomes invalid
B. The open loop, because its prediction is built on the orbit data ✓
C. Neither, because GNSS timing overrides the ephemeris
D. Both equally, because they share the same estimate

Answer: B. The open loop, because its prediction is built on the orbit data

The open loop is pure prediction from the orbit, so bad orbit data corrupts it directly. The closed loop measures the real signal, so it is the stage that partly rescues a stale prediction.

7. What is the stated cost of issuing timing advance commands and AFC updates frequently?

A. The device battery drains faster than the payload can support
B. The satellite must carry a larger soft buffer
C. Uplink and downlink capacity is spent that could otherwise carry data ✓
D. The cyclic prefix must be lengthened for every user

Answer: C. Uplink and downlink capacity is spent that could otherwise carry data

Corrections are signalling, and signalling occupies the same radio resources as user traffic. Correcting more often buys accuracy and pays for it in throughput.

8. Which correction is the network side, rather than the device side, responsible for?

A. Service link Doppler pre-compensation
B. The initial timing advance before the first PRACH
C. The GNSS position fix used in the open loop
D. Feeder link Doppler and the residual measurement ✓

Answer: D. Feeder link Doppler and the residual measurement

The split follows what each end can observe. The device sees its own service link and pre-corrects it. Only the network sees the feeder link and the residual error in the arriving signal.

9. Compared with a terrestrial cell link, the round-trip time on a LEO link is best described as:

A. Long and fast-changing ✓
B. Short and stable
C. Long but essentially constant over a pass
D. Short but highly variable

Answer: A. Long and fast-changing

Both properties matter. Length alone could be absorbed by a fixed offset. It is the fact that the length keeps changing during a pass that forces a continuous correction loop.

10. Which statement captures the design pattern shared by timing advance and frequency offset compensation?

A. Measure first, then predict the next value from the measurement history
B. Predict from known geometry, then refine from real measurement ✓
C. Let the network compute everything and broadcast one common value
D. Avoid prediction entirely and rely on a long cyclic prefix

Answer: B. Predict from known geometry, then refine from real measurement

This is the open-loop then closed-loop pattern stated in the conclusion. Prediction gets the device close enough to be heard at all; measurement closes the remaining gap.

UAV-enhanced 3D beamforming for rural 5G NTN

Group 2 deck, slides 2 to 8

11. Why does a UAV base station need beam steering in elevation, when a terrestrial macro cell largely does not?

A. Elevation steering is required to reduce Doppler on the feeder link
B. The users lie below the platform, so the useful angles are vertical as well as horizontal ✓

C. Regulations cap the azimuth beamwidth of airborne transmitters
D. Elevation steering compensates for the rotation of the Earth

Answer: B. The users lie below the platform, so the useful angles are vertical as well as horizontal

A tower stands among its users, so the interesting variation is horizontal. A UAV hovers above them, so the angle down to a user changes with range, and elevation becomes a real steering axis.

12. What is the main engineering reason for choosing hybrid beamforming rather than fully digital beamforming on a UAV?

A. It gives a wider beam, which tolerates platform tilt better
B. It removes the need for any baseband processing on board
C. It uses fewer RF chains than antenna elements, which saves size, weight and power ✓
D. It allows the UAV to serve users on more than one carrier frequency

Answer: C. It uses fewer RF chains than antenna elements, which saves size, weight and power

A fully digital array needs one RF chain per element. Hybrid beamforming puts a digital precoder behind an analog phase-shift network, so a large array is driven by a small number of chains, which is what an aircraft can lift and power.

13. The control loop runs sense, optimize, act, learn. Which weakness does this loop introduce?

A. It cannot handle users that stay in one place for long periods
B. It requires every user device to run the same learning model
C. It prevents the UAV from changing altitude once deployed
D. Convergence takes time, so link quality dips during fast movement ✓

Answer: D. Convergence takes time, so link quality dips during fast movement

A closed learning loop is always chasing the state it last observed. When the geometry changes faster than the loop converges, the beam is briefly aimed at where the users were rather than where they are.

14. Predictive beam tracking is preferred over repeated exhaustive beam search mainly because:

A. It holds alignment without the overhead of re-searching every beam ✓
B. Exhaustive search cannot find the correct beam at high elevation angles
C. Prediction removes the need for a phase-shift network
D. Exhaustive search is forbidden by the 3GPP NTN specification

Answer: A. It holds alignment without the overhead of re-searching every beam

Sweeping every beam costs time and signalling on every update. Predicting the next weights from the motion already observed reaches the same alignment for a fraction of that overhead.

15. In the multi-tier backhaul, what carries traffic out of the target area?

A. A new fibre spur laid to the nearest terrestrial exchange
B. Inter-UAV mesh links leading up to a HAPS or satellite feeder link ✓
C. A dedicated ground tower built at the edge of the coverage zone
D. Direct device-to-device relaying between the rural user terminals

Answer: B. Inter-UAV mesh links leading up to a HAPS or satellite feeder link

The whole point is to avoid building anything on the ground. UAVs relay to each other and then upward, so no fibre and no tower is needed inside the target area.

16. Altitude is treated as a decision variable rather than a fixed setting. Which trade-off does it control?

A. Transmit power against receiver noise figure
B. Beam count against subcarrier spacing
C. Line-of-sight probability against path loss ✓
D. Backhaul latency against onboard storage

Answer: C. Line-of-sight probability against path loss

Climbing improves the chance of a clear path over terrain and buildings, and at the same time lengthens every link. The best altitude is wherever those two effects balance for the terrain below.

17. Which limitation is described as competing directly with the UAV's endurance?

A. The weight of the antenna array on its own
B. The bandwidth reserved for the feeder link
C. The time taken to compute the digital precoder weights
D. The power drawn by the beamforming hardware and the AI control loop ✓

Answer: D. The power drawn by the beamforming hardware and the AI control loop

On an aircraft the payload and the propulsion draw from one battery. Every watt spent on phase shifters and inference is a watt not spent staying airborne.

18. Backhaul fragility is listed as a risk. What is the specific failure mode?

A. One weak link in the chain degrades coverage for everything downstream of it ✓
B. Two UAVs transmit on the same beam and interfere
C. The satellite feeder link exceeds its power flux density mask
D. The mesh loses time synchronisation and the beams misalign

Answer: A. One weak link in the chain degrades coverage for everything downstream of it

A multi-tier mesh is a chain of dependencies. A UAV that relays through a neighbour inherits that neighbour's problems, so a single weak hop cascades outward.

19. Which of these is given as a real-world application of the approach?

A. Precision timing distribution for financial trading networks
B. Temporary coverage for festivals and other large events ✓
C. Space debris tracking from a UAV-borne radar
D. Inter-continental undersea cable protection monitoring

Answer: B. Temporary coverage for festivals and other large events

The listed cases are rural education and distance learning, temporary event coverage, border surveillance and post-disaster response. All four share a need for coverage that appears and then goes away.

20. The conclusion argues that the binding constraint is system-level. What does that mean in practice?

A. The UAV must be replaced by a HAPS for the system to work
B. Only the weakest single radio link needs to be improved
C. Altitude, beam accuracy, terrain and backhaul have to be optimized together ✓
D. The AI model must be trained on data from the exact deployment site

Answer: C. Altitude, beam accuracy, terrain and backhaul have to be optimized together

Improving one variable in isolation moves the bottleneck rather than removing it. Flying higher helps line of sight and hurts path loss, so the variables have to be solved jointly.

Machine learning for RACH optimization in NTN

Group 3 deck, slides 2 to 12

21. Why is the four-step RACH handshake especially costly over a satellite link?

A. Each of the four messages must be encrypted separately
B. The preamble has to be repeated once per orbital plane
C. The four messages amount to two full satellite round trips before any data moves ✓
D. The base station cannot process message 3 while in view of the user

Answer: C. The four messages amount to two full satellite round trips before any data moves

Messages 1 and 2 are one round trip and messages 3 and 4 are another. On the ground that is about a millisecond. Over a satellite it is tens of milliseconds spent before a single byte of user data has been sent.

22. Differential delay across a narrow LEO beam is about 650 microseconds, and the longest protective cyclic prefix is 684 microseconds. What does that imply?

A. The cyclic prefix is comfortably oversized for LEO operation
B. Differential delay can be ignored because it sits below the prefix length
C. Two preambles must be sent, one for the centre and one for the edge
D. Only about 34 microseconds of margin remains, and wider beams erase it ✓

Answer: D. Only about 34 microseconds of margin remains, and wider beams erase it

The two figures nearly cancel. The guard barely covers the spread today, so any wider beam or higher orbit pushes edge preambles outside the window entirely.

23. What happens when two devices pick the same PRACH preamble?

A. A collision occurs, and both devices must retry one round trip later ✓
B. The network serves the stronger device and silently drops the weaker one
C. The preambles combine and the base station decodes both correctly
D. The base station shortens the backoff window for both devices

Answer: A. A collision occurs, and both devices must retry one round trip later

The preamble is the only thing distinguishing one attempt from another, so identical preambles are indistinguishable. Both attempts fail and both devices pay a full round trip to try again.

24. Doppler over one LEO pass swings from about +48 kHz to about -48 kHz. What does this do to the RACH?

A. The device is barred from transmitting during the sign reversal
B. The preamble is smeared, so correlation at the receiver degrades ✓
C. The random access response arrives on the wrong subcarrier
D. The cyclic prefix is stretched by the frequency change

Answer: B. The preamble is smeared, so correlation at the receiver degrades

Preamble detection is a correlation against a known shape. A frequency offset that moves during the preamble distorts that shape, so the correlation peak drops and detections are missed.

25. In the learning loop, what does the reward signal combine?

A. Beam index, elevation angle and slant range
B. Collision count, subcarrier spacing and preamble length
C. Success, delay and energy ✓
D. Throughput, ephemeris age and satellite battery state

Answer: C. Success, delay and energy

The reward has to express what a good access attempt looks like: it got through, it did not take long, and it did not cost much power. The policy is then tuned to maximise that combination.

26. The predictive or ephemeris-aided family is described differently from the other two. Why?

A. It runs only on the satellite and never on the device
B. It requires far more training data than the other families
C. It has been shown to perform worse than fixed 3GPP parameters
D. It is deterministic geometry, and it is the Release 17 baseline that a learned policy must beat ✓

Answer: D. It is deterministic geometry, and it is the Release 17 baseline that a learned policy must beat

Supervised learning and reinforcement learning both learn from data. Pre-compensating from a known orbit learns nothing; it computes. That makes it the benchmark rather than a competing learned method.

27. Reinforcement learning is applied to which RACH parameters?

A. The barring probability, the backoff window and the preamble pool split ✓
B. The carrier frequency and the subcarrier spacing
C. The cyclic prefix length and the guard period
D. The number of HARQ processes and the soft buffer size

Answer: A. The barring probability, the backoff window and the preamble pool split

These are the contention controls, the values a network operator would otherwise fix at design time. They are exactly the knobs a policy can turn in response to observed load.

28. The reported eRACH result includes a factor of 4.94. What does it measure?

A. The improvement in throughput over fixed rules
B. The increase in collision rate, which is the cost paid for the gain ✓
C. The reduction in access delay
D. The growth in the number of devices that can be served

Answer: B. The increase in collision rate, which is the cost paid for the gain

The gains quoted are +31.2 percent and +54.6 percent throughput and 1.49 times lower delay. The 4.94 figure runs the other way: the learned policy is more aggressive, so it collides more often.

29. Why is heat listed as an onboard limitation separate from power?

A. Heat corrupts the stored neural network weights
B. Thermal noise raises the false preamble detection rate
C. Onboard inference generates heat with no way to vent it in vacuum ✓
D. Heat causes the solar panels to lose efficiency during eclipse

Answer: C. Onboard inference generates heat with no way to vent it in vacuum

On the ground a processor sheds heat into the air. In vacuum there is no convection, so the only path is radiation, and continuous inference produces heat faster than a small payload can radiate it.

30. Which design-process limitation makes regulatory approval difficult?

A. The scarcity of real NTN random-access traces
B. The gap between simulated and real channels
C. The cost of running simulations across many orbital geometries
D. A black-box policy is hard to certify against 3GPP conformance ✓

Answer: D. A black-box policy is hard to certify against 3GPP conformance

The other three are engineering obstacles that better data or more compute would ease. Certification is different: a learned policy has no fixed behaviour to test against a written specification.

GPS signal integration and augmentation in 5G-NTN

Group 4 deck, slides 2 to 16

31. What is the essential difference between DGPS and PPP?

A. DGPS uses carrier phase while PPP uses code measurements only
B. DGPS corrects the ionosphere while PPP corrects the troposphere
C. DGPS works globally while PPP is limited to a single country
D. DGPS needs a nearby surveyed base station, and PPP does not ✓

Answer: D. DGPS needs a nearby surveyed base station, and PPP does not

PPP takes precise orbit and clock products computed by a global monitoring network, so the receiver needs no local reference. DGPS depends on a base station close enough to share the same errors.

32. In DGPS, how is the correction value derived?

A. From the difference between the base station's surveyed position and its GPS-computed position ✓
B. From the carrier phase difference between two satellites in view
C. From a model of ionospheric delay held at the master station
D. From averaging the positions reported by all nearby rovers

Answer: A. From the difference between the base station's surveyed position and its GPS-computed position

The base station already knows exactly where it is. Any disagreement between that truth and what GPS reports is the current error, and that error is what gets broadcast to nearby receivers.

33. Which technique reaches centimetre-level accuracy by using carrier-phase measurements between a base and a rover?

A. SBAS
B. RTK ✓
C. DGPS
D. PPP

Answer: B. RTK

RTK is quoted at 1 to 2 cm. Carrier phase resolves position to a fraction of a wavelength, which is far finer than the code measurements DGPS relies on.

34. In SBAS, what does the geostationary satellite actually do?

A. It measures the GPS errors directly using its own receiver

B. It replaces a failed GPS satellite in the constellation

C. It broadcasts the correction messages that were computed on the ground ✓

D. It relays the user position back to the master control station

Answer: C. It broadcasts the correction messages that were computed on the ground

The errors are detected by surveyed reference stations and turned into correction messages by a master control station. The GEO satellite is only the broadcast channel that carries them over a wide area.

35. A node transmits a positioning signal, the device replies, and the node halves the measured delay. Which technique is this?

A. Time of Arrival

B. Angle of Arrival

C. Trilateration from ephemeris

D. Round Trip Time ✓

Answer: D. Round Trip Time

The halving is the giveaway. Measuring a two-way delay and dividing by two removes the need for the transmitter and the receiver to share a clock, which is what a one-way Time of Arrival measurement would require.

36. Angle of Arrival needs something that Time of Arrival does not. What?

A. An antenna array able to resolve the direction the signal came from ✓

B. A precisely synchronised clock at the transmitter

C. Knowledge of the satellite ephemeris

D. A carrier-phase capable receiver

Answer: A. An antenna array able to resolve the direction the signal came from

AoA works on direction rather than distance, and direction is recovered from the phase differences across the elements of an array. A single antenna cannot tell you where a signal came from.

37. Why is 5G-NTN a useful complement to GPS in urban canyons and indoors?

A. NTN signals are transmitted at a much lower carrier frequency

B. NTN platforms supply extra measurements where the sky view is blocked ✓

C. NTN removes the need for trilateration entirely

D. NTN satellites orbit lower and therefore penetrate buildings

Answer: B. NTN platforms supply extra measurements where the sky view is blocked

GPS fails in those places because too few satellites are visible. Adding ToA, AoA and RTT measurements from NTN platforms restores enough observations to solve for position.

38. SBAS is stated to improve accuracy from what range to what range?

A. From 1 to 3 metres down to about 20 centimetres

B. From 20 metres down to about 5 metres

C. From 5 to 10 metres down to about 1 to 2 metres ✓

D. From 1 to 2 metres down to about 1 to 2 centimetres

Answer: C. From 5 to 10 metres down to about 1 to 2 metres

SBAS sits in the middle of the augmentation ladder. DGPS reaches 1 to 3 m, PPP about 5 to 20 cm after convergence, and RTK 1 to 2 cm.

39. Which statement about the user segment of GPS is correct?

A. It uploads corrected navigation messages to the satellites

B. It monitors satellite health and flags faulty spacecraft

C. It transmits a ranging signal back to the constellation

D. It consists of the receivers and the devices or people using them ✓

Answer: D. It consists of the receivers and the devices or people using them

The user segment only listens. Uploading corrections and monitoring health are control segment jobs, and no GPS receiver transmits anything back to the satellites.

40. Why does the presentation argue that the two systems must be combined?

A. Neither GPS alone nor 5G alone gives good positioning in every condition ✓

B. GPS can provide time but not position

C. 5G-NTN cannot measure distance without GPS assistance

D. Augmentation schemes only work when both systems are present

Answer: A. Neither GPS alone nor 5G alone gives good positioning in every condition

GPS is strong under open sky and weak in canyons, tunnels and indoors. 5G-NTN covers places terrestrial networks cannot. The hybrid exists because the two failure modes do not overlap.

HAPS-based disaster recovery with 5G core integration

Group 5 deck, slides 3 to 11

41. Why can an ordinary unmodified smartphone attach to a HAPS, when it cannot attach directly to most satellites?

A. A HAPS is a standard 5G base station flown at altitude, so the link budget closes to a handset ✓

B. A HAPS transmits on unlicensed spectrum that all phones already support

C. A HAPS uses a modulation scheme designed specifically for handsets

D. Satellites do not implement the 5G NR air interface at all

Answer: A. A HAPS is a standard 5G base station flown at altitude, so the link budget closes to a handset

The deciding factor is distance. At about 20 km the path loss is small enough that a normal phone closes the link, which the deck quotes as roughly 28 dB better than a 500 km LEO.

42. What does the local-breakout UPF on the HAPS payload achieve?

A. It authenticates responders without contacting the ground core

B. Responder traffic can be switched locally and never leaves the disaster zone ✓

C. It stores traffic until the ground network is restored

D. It removes the need for a feeder link to the gateway

Answer: B. Responder traffic can be switched locally and never leaves the disaster zone

The User Plane Function is where user traffic is routed. Putting one on the platform means two responders in the same area can talk through the HAPS alone, without a round trip to the ground core.

43. Which functions stay on the ground 5G core even with a regenerative HAPS payload?

A. Radio resource control and scheduling

B. Local user-plane switching between responders

C. Authentication and session management, and routing to external networks ✓

D. Beamforming weight computation for the service link

Answer: C. Authentication and session management, and routing to external networks

The AMF and the SMF stay on the ground. The payload carries the gNB and a breakout UPF, so it handles the radio and local traffic, but subscriber authentication and external routing remain a core function.

44. How does HAPS end-to-end latency compare with GEO in the quoted table?

A. Roughly the same, since both are line-of-sight links

B. HAPS is around one order of magnitude lower

C. HAPS is higher because of the extra gateway hop

D. HAPS is roughly fifty times lower ✓

Answer: D. HAPS is roughly fifty times lower

The table gives 1 to 10 ms for HAPS against 480 to 560 ms for GEO. Comparing the middle of each range puts the ratio near fifty.

45. What link-budget advantage is quoted for a 20 km HAPS over a 500 km LEO for direct-to-smartphone service?

A. About 28 dB ✓

B. About 3 dB

C. About 12 dB

D. About 60 dB

Answer: A. About 28 dB

The figure follows from the range ratio. Being twenty-five times closer buys roughly 28 dB of path loss, which is what makes an unmodified handset viable.

46. Which challenge belongs to the Ka-band feeder link rather than the service link?

A. Station-keeping against stratospheric winds
B. Rain fade, which has to be handled with site diversity ✓
C. Airspace clearance for the platform
D. Sharing one platform's capacity across the whole footprint

Answer: B. Rain fade, which has to be handled with site diversity

Rain attenuation grows sharply with frequency, so a 20 to 30 GHz feeder link is vulnerable in a way the lower-frequency service link is not. Site diversity answers it by having a second gateway in different weather.

47. A single HAPS covers a 50 to 100 km radius. What does that mean for capacity?

A. Capacity scales automatically with the radius covered
B. The radius must be reduced before the data rate can rise
C. One platform's capacity is shared across that entire footprint ✓
D. Coverage beyond 50 km requires a second feeder link

Answer: C. One platform's capacity is shared across that entire footprint

Coverage and capacity are different quantities. A wide footprint is an advantage for reach and a disadvantage for throughput per user, because every user inside it draws on the same payload.

48. Why is deployment speed the decisive advantage of HAPS in disaster response?

A. The platform can be recovered and reused after the event
B. It needs no gateway on the ground
C. It operates in spectrum permanently reserved for emergencies
D. It can be flown in hours or days, with no launch campaign ✓

Answer: D. It can be flown in hours or days, with no launch campaign

A satellite has to be built, scheduled and launched. Disaster response is measured in hours, and a platform that can be in the air the same day is the only one that arrives while it still matters.

49. Endurance is listed as an open problem. What figure is quoted?

A. 12 days is the current airship record ✓
B. 12 hours is the current airship record
C. 12 weeks is the current airship record
D. 12 months is the current airship record

Answer: A. 12 days is the current airship record

Twelve days is long enough for an emergency deployment and far short of permanent infrastructure, which is why the deck calls HAPS a temporary layer by design.

50. The conclusion attaches a condition to the claim that HAPS restores coverage. What is it?

A. The terrestrial towers must be only partially damaged
B. Backhaul must survive through a gateway, a satellite or another HAPS ✓
C. The handsets must be upgraded to support NTN
D. The disaster zone must be smaller than 50 km across

Answer: B. Backhaul must survive through a gateway, a satellite or another HAPS

A HAPS restores the radio access network, not the transport behind it. Without a surviving path to the core, the platform can connect phones to each other locally but not to anything beyond the zone.

Multi-connectivity and session continuity across TN-NTN links

Group 6 deck, slides 3 to 19

51. What distinguishes make-before-break handover from a conventional handover?

A. The old link is released before the new one is established
B. The new link is established before the old one is released ✓
C. Both links are released and the device re-attaches from idle

D. The handover is decided by the device rather than the network

Answer: B. The new link is established before the old one is released

A conventional handover has a gap between releasing one link and acquiring the next. Overlapping the two removes that gap, which is what stops the session from dropping.

52. A device is attached to a 5G base station and a LEO satellite at the same time. Which technology enables that?

A. Access Traffic Steering, Switching and Splitting
B. Network slicing
C. Multi-Radio Dual Connectivity ✓
D. Multi-Access Edge Computing

Answer: C. Multi-Radio Dual Connectivity

MR-DC is the mechanism that lets one device hold simultaneous connections over two different radio accesses. ATSSS then decides what traffic to send over which of them.

53. ATSSS and MR-DC are related but distinct. What does ATSSS add?

A. It creates the simultaneous radio connections in the first place
B. It authenticates the device on each access network separately
C. It predicts when a handover will become necessary
D. It decides how traffic is routed and balanced across the connections ✓

Answer: D. It decides how traffic is routed and balanced across the connections

MR-DC provides the pipes and ATSSS provides the policy. Steering, switching and splitting are three ways of dividing a traffic flow across pipes that already exist.

54. Which challenge is matched with intelligent routing as its solution?

A. Latency differences between the access types ✓
B. Frequent handovers
C. Link failures
D. Session interruption

Answer: A. Latency differences between the access types

A satellite path and a terrestrial path have very different delays. Routing intelligently means sending delay-sensitive traffic down the short path and bulk traffic down the long one.

55. Why does MEC help specifically during mobility?

A. It removes the need for a satellite feeder link
B. It processes data closer to the user, so latency stays low as the link changes ✓
C. It stores the session context so the device need not re-authenticate
D. It selects which satellite the device should connect to next

Answer: B. It processes data closer to the user, so latency stays low as the link changes

Edge computing shortens the path between the user and the application. When the access link is switching between terrestrial and satellite, keeping the compute nearby limits how much the round trip can grow.

56. How is session continuity defined?

A. A device stays attached to one satellite for a whole pass
B. A fixed data rate is reserved for the duration of a call
C. An ongoing session is preserved while the underlying link changes ✓
D. Both terrestrial and satellite links are always kept available

Answer: C. An ongoing session is preserved while the underlying link changes

Continuity is about the session surviving, not about the link staying still. The benefits listed follow from that: no repeated logins, no data loss and a stable experience.

57. How does network slicing contribute to seamless connectivity?

A. It reduces the number of handovers a device must perform
B. It splits a single session across two carriers to raise throughput
C. It pre-computes the beam weights for the satellite link
D. It gives each application a dedicated virtual network with consistent quality of service ✓

Answer: D. It gives each application a dedicated virtual network with consistent quality of service

Slicing isolates services from one another, so a latency-critical application keeps its guarantees even when the network beneath it changes access type or comes under load.

58. Which benefit belongs to multi-connectivity rather than to session continuity?

- **A. Load balancing and a backup path if one link fails ✓**
- B. No repeated logins
- C. No data loss during a link change
- D. A stable experience for the end user

Answer: A. Load balancing and a backup path if one link fails

Load balancing and redundancy require two links running at once, which is what multi-connectivity provides. The other three are what continuity delivers once the link does change.

59. Frequent handovers are listed as an NTN challenge. What causes them?

- A. The high altitude of the satellite platform
- **B. LEO satellites sweep across the sky, so the serving cell keeps changing ✓**
- C. Congestion in the terrestrial network
- D. The mismatch in latency between the two access types

Answer: B. LEO satellites sweep across the sky, so the serving cell keeps changing

In a terrestrial network the user moves past fixed towers. In a LEO network the tower itself moves, so even a stationary user is handed over every few minutes.

60. AI-assisted mobility management turns the handover from reactive into proactive. How?

- A. By raising the signal strength threshold that triggers a handover
- B. By handing over to the satellite whenever the terrestrial link weakens
- **C. By predicting user movement and network conditions ahead of time ✓**
- D. By keeping the device on the terrestrial link for as long as possible

Answer: C. By predicting user movement and network conditions ahead of time

A reactive handover waits for a measurement to cross a threshold. Predicting the movement lets the network prepare the target in advance, so the switch happens before the link degrades.

Spectrum sharing and interference management between 5G NTN and TN

Group 7 deck, slides 3 to 14

61. A PFD mask is expressed as a function of elevation angle. Why does the elevation angle matter?

- A. The satellite transmits more power at low elevation
- B. Doppler shift is largest at high elevation angles
- **C. It determines how much satellite energy reaches a terrestrial receiver at the surface ✓**
- D. The terrestrial base station antenna tilts with elevation

Answer: C. It determines how much satellite energy reaches a terrestrial receiver at the surface

A beam arriving steeply from overhead spreads its energy differently from one arriving near the horizon, and terrestrial antennas discriminate against those angles differently. The regulatory limit therefore has to be angle dependent.

62. What does EPFD add beyond an ordinary PFD limit?

- A. It applies the limit to the uplink as well as the downlink
- B. It converts the limit into a time-averaged rather than a peak value
- C. It extends the limit to adjacent bands as well as the co-channel band
- **D. It accounts for the aggregate interference from a whole constellation ✓**

Answer: D. It accounts for the aggregate interference from a whole constellation

A single spacecraft can comply while a constellation of hundreds, all visible at once, still swamps a terrestrial receiver. EPFD is the equivalent limit applied to the summed contribution.

63. Null-steering suppresses interference without simply reducing transmit power. How?

- **A. It places a null in the radiation pattern toward the victim receiver ✓**
- B. It narrows the main lobe until it misses the victim receiver

C. It shifts the beam into an adjacent frequency band
D. It reduces the number of active antenna elements

Answer: A. It places a null in the radiation pattern toward the victim receiver

An adaptive array can shape its pattern, not just point it. Steering a null at the victim removes energy in that one direction while the main lobe keeps serving its intended users at full power.

64. Which interference mechanism arises purely from the speed of LEO satellites?

A. Co-channel interference
B. Doppler-induced spreading ✓
C. Adjacent-band leakage
D. Cross-border coexistence

Answer: B. Doppler-induced spreading

The other three exist for a stationary satellite too. Only spectral smearing from a rapidly changing frequency offset is a direct consequence of orbital velocity.

65. In the uplink interference scenario, who is the victim?

A. A terrestrial user equipment inside the satellite footprint
B. The terrestrial base station, reached by the satellite downlink
C. The satellite receiver, reached by a terrestrial user's transmission ✓
D. The gateway earth station on the feeder link

Answer: C. The satellite receiver, reached by a terrestrial user's transmission

Direction defines the victim. On the downlink the satellite beam spills into the terrestrial cell and a ground user suffers. On the uplink the terrestrial transmission rises into the satellite and the satellite suffers.

66. Which approach sits underneath dynamic spectrum sharing rather than beside it?

A. Exclusive or licensed allocation
B. Static co-primary sharing
C. Geographic separation using exclusion zones
D. Cognitive sensing ✓

Answer: D. Cognitive sensing

The tree has four top-level approaches. Dynamic spectrum sharing then branches into power control, adaptive beamforming, cognitive sensing and AI/ML-based allocation.

67. Which 3GPP release is identified as the first NTN specification?

A. Release 17 ✓
B. Release 15
C. Release 18
D. Release 20

Answer: A. Release 17

Releases 15 and 16 were study items. Release 17 in 2022 is where NR-NTN and NB-IoT or eMTC over satellite were first specified rather than merely studied.

68. Why is cross-border regulation listed as an open challenge?

A. Satellites cannot switch frequency as they cross a border
B. Different countries have to agree on the same PFD rules ✓
C. Terrestrial operators are not licensed to serve foreign users
D. Doppler shift changes as the satellite crosses a boundary

Answer: B. Different countries have to agree on the same PFD rules

A beam does not stop at a national boundary, but spectrum regulation does. Two neighbouring administrations with different limits leave the operator with no single rule to comply with.

69. When are guard bands and guard time slots used?

A. When the satellite must reduce its power flux density
B. When the constellation exceeds its EPFD budget
C. When co-channel sharing between NTN and TN is not workable ✓
D. When the terrestrial network is congested

Answer: C. When co-channel sharing between NTN and TN is not workable

Guard resources are the fallback. They separate the two systems in frequency or in time and give up the efficiency that sharing the same resources would have provided.

70. The takeaways name one direction as the clear road to 6G. Which?

A. Exclusive licensed allocation for NTN
B. Wider guard bands between NTN and TN
C. Moving all NTN traffic into Ka-band
D. AI/ML-driven dynamic spectrum sharing ✓

Answer: D. AI/ML-driven dynamic spectrum sharing

The other three all work by keeping the systems apart, which wastes spectrum. Learning to share the same spectrum in real time is what the deck identifies as the path forward.

Mobility management and handover optimization in 5G NTN

Group 8 deck, slides 4 to 14

71. Which RRC state keeps the UE context stored in the network while the connection itself is released?

A. RRC CONNECTED
B. RRC IDLE
C. RRC SETUP
D. RRC INACTIVE ✓

Answer: D. RRC INACTIVE

INACTIVE is the middle state and it exists to make resuming cheap. IDLE keeps no context, so returning to CONNECTED from IDLE needs a full RRC setup.

72. Why is a purely signal-strength-based handover trigger inadequate in NTN?

A. The satellite keeps moving, so the measurement is out of date by the time it is acted on ✓
B. Satellites do not transmit reference signals that can be measured
C. Signal strength cannot be measured through the ionosphere
D. The device has no way to report measurements on the uplink

Answer: A. The satellite keeps moving, so the measurement is out of date by the time it is acted on

A measurement report has to travel to the network and a decision has to travel back. Over a satellite link that takes long enough for the geometry that justified the report to have changed.

73. A time-based trigger fires at a predicted moment. Where does the prediction come from?

A. The historical measurement log stored in the device
B. Satellite ephemeris and orbit data ✓
C. The current network load at the target satellite
D. The rate of change of the received signal strength

Answer: B. Satellite ephemeris and orbit data

Orbits are deterministic, so the time at which a given satellite will fall below a usable elevation can be computed in advance rather than waited for.

74. What distinguishes conditional handover from a location-based trigger on its own?

A. It uses no location information at all
B. It is executed by the device without network involvement
C. Several conditions have to be satisfied together before it fires ✓
D. It always hands over to a beam on the same satellite

Answer: C. Several conditions have to be satisfied together before it fires

Location is one of the conditions in the set, alongside link quality, time to degradation and target load. Requiring all of them at once is what cuts unnecessary handovers.

75. Which condition in the conditional handover set concerns the target rather than the radio link?

A. Target SINR exceeds serving SINR plus a hysteresis margin
B. Time to link degradation is below a threshold

C. The UE is inside the target coverage area

D. The target satellite's load is below a maximum ✓

Answer: D. The target satellite's load is below a maximum

Load is a capacity property of the target cell, not a property of the radio path. Checking it stops the network from handing a device onto a satellite that is already full.

76. How does beam-based handover differ from satellite handover?

A. The device switches between spot beams of the same satellite ✓

B. The device has to change carrier frequency

C. It is triggered only by time and never by location

D. It removes the need for a timing advance update

Answer: A. The device switches between spot beams of the same satellite

A single satellite projects many spot beams, and the footprint of each sweeps across the ground. A user can leave one beam and enter another without ever changing satellite.

77. Reserving resources at the target satellite before a handover carries which specific cost?

A. The device battery drains from continuous location fixing

B. Capacity is wasted whenever the handover does not actually take place ✓

C. Doppler pre-compensation becomes impossible

D. The RRC state has to fall back to IDLE

Answer: B. Capacity is wasted whenever the handover does not actually take place

Preparing in advance means committing resources on a prediction. Every prediction that does not come true has held capacity idle that could have served someone else.

78. Why does a location-based trigger drain the device battery?

A. It requires the device to stay in RRC CONNECTED at all times

B. It doubles the number of transmit chains in use

C. The device has to keep obtaining position fixes ✓

D. It forces the device to measure every visible satellite

Answer: C. The device has to keep obtaining position fixes

Knowing when a geographic boundary has been crossed means knowing where you are continuously. A GNSS receiver running all the time is a significant, constant power draw.

79. Make-before-break is described as adding hardware complexity. Why?

A. The device has to store the full ephemeris of the constellation

B. The satellite has to carry an additional soft buffer

C. The device has to run an AI model in real time

D. The device has to support two simultaneous links ✓

Answer: D. The device has to support two simultaneous links

Holding the old link while the new one comes up means two active radio connections at once, which needs the receive and transmit hardware to support both.

80. How does the conclusion characterise the overall shift in NTN mobility management?

A. From reactive and signal-based to predictive and ephemeris-driven ✓

B. From ephemeris-driven to measurement-driven

C. From network-controlled to device-controlled

D. From beam-based to satellite-based handover

Answer: A. From reactive and signal-based to predictive and ephemeris-driven

The old model waits for a measurement to degrade. The new one exploits the fact that satellite motion is known in advance, and pays for that with signalling, battery and reserved capacity.

ISAC-enabled non-terrestrial networks for 6G

Group 9 deck, slides 2 to 11

81. What is the defining feature of Integrated Sensing and Communication?

A. One waveform, one RF front end and one slice of spectrum serve both functions ✓
B. A radar payload is flown alongside a communication payload on one satellite
C. Sensing data is carried as payload traffic over the communication link
D. A communication link switches between data and radar modes in turn

Answer: A. One waveform, one RF front end and one slice of spectrum serve both functions

The integration is in the hardware and the waveform, not in the packaging. Two payloads on one bus, or time-sharing one link, both keep the functions separate; ISAC does not.

82. Why is a GEO satellite ranged rather than tracked by Doppler?

A. GEO satellites transmit at a frequency too high for Doppler processing
B. GEO shows almost no Doppler because it is nearly stationary relative to the ground ✓
C. GEO Doppler is masked by ionospheric scintillation
D. GEO orbits are known exactly, so no tracking is needed at all

Answer: B. GEO shows almost no Doppler because it is nearly stationary relative to the ground

Doppler needs relative radial motion. A geostationary satellite holds its position over a point on the ground, so the shift is close to zero and there is no signature to track.

83. In a multistatic ISAC configuration, what is the arrangement?

A. Each satellite transmits and listens for its own echo
B. One satellite transmits and a single second satellite receives the reflection
C. One transmission is received by several nodes across the constellation ✓
D. The ground station transmits and the satellites all receive

Answer: C. One transmission is received by several nodes across the constellation

The deck shows one transmitter and receivers Rx1 through Rx4 all catching the same echo. Many viewing angles on one target is what turns the constellation itself into a radar.

84. Self-interference is a problem in which geometry, and why?

A. Bistatic, because the two nodes are not synchronised
B. Multistatic, because several echoes arrive at once
C. All three equally, because the waveform is shared
D. Monostatic, because the receiver must hear a faint echo through its own loud transmitter ✓

Answer: D. Monostatic, because the receiver must hear a faint echo through its own loud transmitter

Sharing one aperture means transmitting and receiving at the same time and place. The returning echo is many orders of magnitude weaker than the outgoing signal sitting on top of it.

85. Roughly how much cancellation is quoted as necessary for the monostatic case?

A. More than 70 dB ✓
B. More than 20 dB
C. More than 40 dB
D. More than 120 dB

Answer: A. More than 70 dB

The deck specifies more than 70 dB of combined analog and digital cancellation, and notes that it has to be achieved on a moving payload, which makes it harder still.

86. What is the principal challenge that bistatic and multistatic operation introduces, and that monostatic avoids?

A. Self-interference between transmit and receive
B. Time and phase synchronisation across independent nodes ✓
C. The need for a joint waveform design
D. The absence of a native ISAC standard

Answer: B. Time and phase synchronisation across independent nodes

One aperture shares one clock by construction. Splitting transmit from receive across separate spacecraft means those spacecraft have to agree on time and phase to sub-microsecond accuracy.

87. Why does limited onboard power and compute add latency to ISAC over NTN?

A. The waveform has to be transmitted at a lower rate
B. The receiver must integrate over a longer dwell time

C. Processing is pushed down to the ground rather than done on board ✓
D. The satellite must store echoes until it passes a gateway

Answer: C. Processing is pushed down to the ground rather than done on board

If the payload cannot process the returns itself, the raw data has to be downlinked and processed on the ground, and the answer comes back a full round trip later.

88. Which advantage follows directly from not needing dedicated sensing spectrum?

A. High-precision localization
B. Extreme disaster resilience
C. Real-time environment mapping
D. Enhanced spectral efficiency and reduced congestion ✓

Answer: D. Enhanced spectral efficiency and reduced congestion

Conventional radar occupies a band that carries no data. Folding the sensing into the communication waveform frees that allocation, which is a spectrum-efficiency gain rather than a capability gain.

89. In the maritime use case, what is meant by one beam doing two jobs?

A. The beam tracks ships and reads sea state while also serving broadband ✓
B. The beam serves two vessels at the same time
C. The beam alternates between uplink and downlink
D. The beam covers both the coastline and the open ocean

Answer: A. The beam tracks ships and reads sea state while also serving broadband

The two jobs are sensing and communication, not two users or two directions. Over water there are no towers, so the same beam has to earn its place twice.

90. DebriSense-THz is quoted with which performance figure?

A. 70 dB of interference cancellation at 5 THz
B. 95 to 99 percent classification accuracy at 5 THz ✓
C. Sub-microsecond synchronisation at 5 THz
D. 95 to 99 percent spectral efficiency gain at 5 THz

Answer: B. 95 to 99 percent classification accuracy at 5 THz

The figure belongs to the space situational awareness use case: detecting, tracking and classifying orbiting debris, where classification accuracy is the metric that matters.

AI-native Open RAN for NTN

Group 10 deck, slides 3 to 13

91. What separates the Non-RT RIC from the Near-RT RIC?

A. The Non-RT RIC runs on the satellite and the Near-RT RIC on the ground
B. The Non-RT RIC runs loops slower than one second and the Near-RT RIC faster than one second ✓
C. The Non-RT RIC handles user data and the Near-RT RIC handles control signalling
D. The Non-RT RIC is vendor-specific and the Near-RT RIC is open

Answer: B. The Non-RT RIC runs loops slower than one second and the Near-RT RIC faster than one second

The split is by control-loop timescale. Slow work such as policy and model training goes above one second; fast optimisation through xApps goes below it.

92. Which functions sit in the O-CU-CP?

A. RLC and MAC
B. SDAP and PDCP
C. RRC and PDCP ✓
D. High-PHY and Low-PHY

Answer: C. RRC and PDCP

The control-plane half of the central unit carries RRC together with PDCP. SDAP with PDCP is the user-plane half, and RLC with MAC belongs to the distributed unit.

93. What is the main structural difference between traditional RAN and Open RAN?

A. Open RAN removes the need for a core network
B. Open RAN places all processing in the radio unit
C. Open RAN uses satellites instead of ground towers
D. Open RAN splits the chain into functions that may come from different vendors ✓

Answer: D. Open RAN splits the chain into functions that may come from different vendors

A traditional RAN is a single vendor's baseband unit and radio unit joined by proprietary interfaces. Open RAN standardises those interfaces so O-CU, O-DU and O-RU can be sourced separately.

94. In the AI-RAN model, which layer covers AI hardware, AI software and AI integration used to improve the network itself?

A. AI-for-RAN ✓
B. AI-on-RAN
C. AI-and-RAN
D. AI-over-RAN

Answer: A. AI-for-RAN

The three layers are concentric. AI-for-RAN is the innermost and points inward at the network. AI-and-RAN shares infrastructure, and AI-on-RAN sells services outward over the network.

95. AI-on-RAN is best described as:

A. Using AI to optimize scheduling inside the base station
B. Delivering AI services to customers over the network ✓
C. Sharing infrastructure between AI workloads and RAN workloads
D. Training AI models on the satellite payload

Answer: B. Delivering AI services to customers over the network

It is the outermost layer, covering the marketplace platform, developer tools and global access. The network becomes the delivery channel for AI rather than the thing being optimised.

96. Which component actually allocates satellite beams and resources to user devices in the described NTN pipeline?

A. The Service Management and Orchestration framework
B. rApps running on the Non-RT RIC
C. xApps running on the Near-RT RIC ✓
D. The O-RU on board the satellite

Answer: C. xApps running on the Near-RT RIC

The machine learning engine predicts and optimises, but the allocation itself is a fast control action, and fast control actions are what xApps on the Near-RT RIC exist to perform.

97. Why does satellite mobility create a problem for Open RAN specifically?

A. Open interfaces cannot carry ephemeris data
B. The O-Cloud cannot be deployed on a moving platform
C. xApps must be recompiled for each orbital plane
D. O-RAN interfaces assume a static fronthaul with predictable latency ✓

Answer: D. O-RAN interfaces assume a static fronthaul with predictable latency

The functional splits were designed for a fibre run of known length between a baseband unit and a radio unit. A moving satellite changes that path length continuously, which breaks the timing assumption.

98. The RIC placement question in NTN is a choice between:

A. The ground segment and the satellite ✓
B. The core network and the radio unit
C. The Non-RT RIC and the Near-RT RIC
D. A single vendor and multiple vendors

Answer: A. The ground segment and the satellite

Putting the RIC on the ground keeps compute cheap and adds a round trip to every decision. Putting it on the satellite removes the round trip and spends scarce onboard power.

99. Which AI application is described as using signal strength, network load and user motion

together?

A. Beam optimization

B. Intelligent handover ✓

C. Self-optimizing network operations

D. Functional split selection

Answer: B. Intelligent handover

Those three inputs feed the handover decision, so the network can choose the best target satellite along an intelligent prediction path rather than reacting to one measurement.

100. The claim that resource optimization gets harder after deployment rests on which fact?

A. The constellation grows over time as more satellites are launched

B. The AI model degrades as its training data ages

C. The geometry keeps changing once the constellation is flying ✓

D. Open interfaces cannot be updated after deployment

Answer: C. The geometry keeps changing once the constellation is flying

A terrestrial network is planned once and then largely holds still. A constellation never holds still, so the optimisation problem is re-posed continuously rather than solved at planning time.

Federated learning for CSI feedback and beam management in LEO NTN

Group 11 deck, slides 2 to 12

101. What does channel state information tell the satellite, and why does it have to be reported?

A. Which satellite the device intends to hand over to next

B. The device's GNSS position, so the beam can be pointed at it

C. How the radio path changes the signal, which the satellite needs before it can aim a beam ✓

D. How much buffer the device has left for incoming data

Answer: C. How the radio path changes the signal, which the satellite needs before it can aim a beam

CSI describes the channel in strength and timing. The satellite cannot observe the downlink channel itself, so the device has to measure it and report it back before precoding is possible.

102. In the conventional cycle the device measures, quantises, reports and the gNB precodes. What is the stated cost of that cycle?

A. Each report has to be acknowledged by the satellite before the next one

B. The device has to re-authenticate at the start of every reporting cycle

C. Quantisation forces the device to transmit at maximum power

D. The report must be repeated often, and its cost grows with antennas, users and report rate ✓

Answer: D. The report must be repeated often, and its cost grows with antennas, users and report rate

The report is small on its own but it is sent constantly, by every user, and it grows with the size of the antenna array. Multiplied out, it becomes the dominant uplink load.

103. Challenge one is that the report is stale on arrival. What makes it stale?

A. The satellite moves at about 7.5 km/s, so the channel changes between measurement and use ✓

B. The report is queued behind higher-priority traffic in the uplink buffer

C. The codebook entry has to be looked up on the ground before use

D. The device only measures the channel once per connection

Answer: A. The satellite moves at about 7.5 km/s, so the channel changes between measurement and use

The deck lays the timeline out as measure, send, arrive, applied. The interval over which the channel stays valid is shorter than the measure-to-use delay, so the report describes a channel that has gone.

104. Challenge two is that the data needed to fix the problem is private. Why is CSI private?

A. CSI reports include the subscriber identity in the header

B. A channel measurement is nearly unique to one place, so it identifies where a user is ✓
C. The codebook index reveals which services the user is running
D. CSI is transmitted without ciphering on the uplink

Answer: B. A channel measurement is nearly unique to one place, so it identifies where a user is

The deck shows a measured CSI grid being turned back into a position. The multipath signature of a location acts as a fingerprint, so collecting raw CSI centrally exposes subscriber location.

105. What does the learned encoder achieve, and what compression ratios are quoted?

A. It removes the need for any uplink report at all
B. It encrypts the CSI so the aggregator cannot read it, at ratios from 1/2 to 1/8
C. It compresses the CSI into a short codeword, at ratios from 1/16 to 1/64 ✓
D. It predicts the next CSI report, at ratios from 1/4 to 1/16

Answer: C. It compresses the CSI into a short codeword, at ratios from 1/16 to 1/64

An autoencoder is trained as a pair: an encoder on the device and a decoder at the gateway. Only the codeword crosses the link, which is where the 1/16 to 1/64 saving comes from.

106. Why must the encoder and the decoder be trained together as one pair?

A. Training them separately would exceed the device's memory budget
B. 3GPP requires joint training for all air-interface models
C. Separate training would make the codeword longer than the raw report
D. The decoder has to learn to invert the specific representation that the encoder produces ✓

Answer: D. The decoder has to learn to invert the specific representation that the encoder produces

The codeword is not a standard format. It is whatever internal representation the encoder settles on, so only a decoder trained alongside it can reconstruct the channel from it.

107. What is the defining property of federated learning in this design?

A. Model updates cross the link while the raw CSI never leaves the device ✓
B. Each device trains a completely separate model that is never combined
C. Training happens only on the satellite, using data it collects itself
D. The global model is broadcast but never updated after deployment

Answer: A. Model updates cross the link while the raw CSI never leaves the device

Devices train locally on their own data and send only the resulting updates. The aggregator averages those updates, so it improves a shared model without ever holding anyone's raw measurements.

108. Which algorithms are named as what the aggregator runs?

A. DQN, DDPG, PPO and SAC
B. FedAvg, FedProx, SCAFFOLD and FedAdam ✓
C. FedAvg, Q-learning, LSTM and SGD
D. AES, Kyber, Dilithium and SHA-256

Answer: B. FedAvg, FedProx, SCAFFOLD and FedAdam

These are federated aggregation methods: FedAvg is the baseline, FedProx handles uneven data, SCAFFOLD corrects drift and FedAdam is an adaptive server. The other lists are reinforcement learning and cryptography.

109. Where does putting the aggregator on the satellite sit in the trade-off?

A. It is the slowest option but it uses no onboard power
B. It removes the need for secure aggregation
C. It is the fastest option and it costs onboard power ✓
D. It stops the satellite from being visible for longer than a few minutes

Answer: C. It is the fastest option and it costs onboard power

Ground-assisted aggregation makes every round cross the whole link. Aggregating on the satellite removes that trip, which is why it is fastest, and spends scarce payload power doing it.

110. Applying the same idea to beam management replaces an exhaustive sweep with what?

A. A fixed beam schedule computed once at deployment
B. A request to the user device to nominate its preferred beam
C. A random beam selection that is corrected by the closed loop

D. A predictor fed by ephemeris and past beams that outputs the best beam directly ✓

Answer: D. A predictor fed by ephemeris and past beams that outputs the best beam directly

In the sweep the device measures every beam and reports the best. With a predictor it measures once to confirm the prediction, which is where the reporting overhead is saved.

AI-driven dynamic beam control for LEO 5G-NTN

Group 12 deck, slides 3 to 14

111. Why does a fixed beam plan go out of date within seconds in a LEO network?

A. The user devices move too fast for a fixed plan to track
B. The subcarrier spacing changes as the elevation angle changes
C. Beam weights are erased whenever the satellite passes into eclipse
D. The cell itself moves at orbital speed, so the footprint has shifted ✓

Answer: D. The cell itself moves at orbital speed, so the footprint has shifted

In a terrestrial network the cell is fixed and the users move. In LEO the reverse dominates: the beam footprint sweeps the ground at kilometres per second, so any plan set in advance is stale immediately.

112. The deck shows some cells at high demand and others at low or no demand under a uniform power allocation. What problem does this illustrate?

A. Highly dynamic and uneven traffic wastes resources on empty cells ✓
B. Co-channel interference between adjacent beams
C. Timing misalignment between the SSB and the PRACH
D. Doppler spreading across the beam footprint

Answer: A. Highly dynamic and uneven traffic wastes resources on empty cells

Applying the same power to all cells means the busy cells are underserved while the empty ones consume payload resources that produce nothing.

113. In the scheduling conflict shown, two cells both need the beam during mandatory signalling. What is the consequence?

A. Both cells receive the beam at half power
B. One of the cells is left with a coverage hole ✓
C. The satellite defers the signalling to the next orbit
D. The devices in both cells fall back to a terrestrial carrier

Answer: B. One of the cells is left with a coverage hole

SSB and PRACH occupancy is fixed by the standard, so the beam cannot serve both cells at that instant. Whichever cell loses the contest is not covered when it needs to be.

114. Under spatial-domain beam control, what is user-beam mapping for?

A. Assigning each user a unique preamble within the beam
B. Deciding which users are handed over to the next satellite
C. Grouping users by location so co-channel interference across adjacent beams is minimised ✓
D. Matching each user to a network slice

Answer: C. Grouping users by location so co-channel interference across adjacent beams is minimised

How users are grouped into beams decides which of them end up sharing a frequency with a neighbouring beam. Grouping by location keeps reuse distances sensible.

115. What is footprint shaping intended to compensate for?

A. Rain fade on the feeder link
B. The reversal of Doppler sign at the zenith
C. Uneven battery discharge across the payload
D. Changing slant range as the satellite moves from horizon to zenith ✓

Answer: D. Changing slant range as the satellite moves from horizon to zenith

A beam of fixed angular width covers a much larger and more distorted ground area near the horizon than overhead. Widening or narrowing it keeps the covered area consistent through the pass.

116. Beam hopping is a time-domain technique. What does it do?

- **A. It time-multiplexes physical beams across several ground cells to serve sporadic demand ✓**
- B. It moves a beam to a new frequency when interference is detected
- C. It switches a user between two satellites without a handover
- D. It reshapes the beam as the slant range changes

Answer: A. It time-multiplexes physical beams across several ground cells to serve sporadic demand

With fewer beams than cells, the way to serve them all is to visit each in turn. Cells with more traffic are visited more often, which is how sporadic demand is met efficiently.

117. What is signalling alignment concerned with?

- A. Aligning the beam boresight with the user's GNSS position
- **B. Reserving the 3GPP-defined slots for synchronisation and broadcast so they do not collide with data ✓**
- C. Matching the uplink and downlink frame timing at the gateway
- D. Synchronising the phase of the elements in the antenna array

Answer: B. Reserving the 3GPP-defined slots for synchronisation and broadcast so they do not collide with data

Signals such as the SSB have to appear at defined times. The hopping schedule must work around those reservations rather than through them.

118. In the 3GPP AI execution models, what characterises Type 3?

- A. Joint training at one side only
- B. Joint training at two sides using forward activation and backward gradient
- **C. Separate training at two sides, with a training dataset shared between them ✓**
- D. Training performed entirely on the satellite payload

Answer: C. Separate training at two sides, with a training dataset shared between them

Type 1 trains at one side. Type 2 trains jointly across both sides by exchanging activations and gradients. Type 3 keeps the training separate and shares a dataset instead.

119. Which limitation is stated as making a beam prediction error highly visible?

- A. A wrong prediction raises the collision rate on the RACH
- B. A wrong prediction forces the satellite into a safe mode
- C. A wrong prediction increases the reporting overhead on the uplink
- **D. An incorrect prediction cuts coverage completely and causes a visible outage ✓**

Answer: D. An incorrect prediction cuts coverage completely and causes a visible outage

Beams are narrow and directional. Aiming one at the wrong place does not merely degrade service in that cell, it removes it, and the users there see a complete outage.

120. What does the conclusion identify as the central engineering trade-off of AI-driven beam control?

- **A. Efficiency is bought with onboard computing power ✓**
- B. Coverage is bought with additional spectrum
- C. Latency is bought with a longer cyclic prefix
- D. Reliability is bought with a larger constellation

Answer: A. Efficiency is bought with onboard computing power

Steering beams where capacity is actually needed raises efficiency, and the prediction that makes it possible has to run on a payload with a hard power and thermal budget.

GPS and Galileo

Group 13 deck, slides 3 to 31

121. A GNSS receiver needs a minimum of four satellites rather than three. What does the fourth one resolve?

- **A. The receiver's own clock bias ✓**
- B. The ambiguity between two candidate points in space

C. The ionospheric delay along each path
D. The identity of the constellation being tracked

Answer: A. The receiver's own clock bias

Three ranges fix a point in three dimensions if the clocks agree. A consumer receiver has no atomic clock, so its offset is a fourth unknown and needs a fourth equation.

122. Why is a pseudorange called pseudo?

A. It is measured to a predicted satellite position rather than the real one
B. It contains the receiver's time-synchronisation error, so it is not the true physical distance ✓
C. It is derived from carrier phase rather than from the ranging code
D. It is an average over several satellites rather than a single measurement

Answer: B. It contains the receiver's time-synchronisation error, so it is not the true physical distance

The receiver converts travel time into distance using its own imperfect clock. That clock error appears in the result as an offset common to every satellite, which is what the fourth measurement removes.

123. A one microsecond clock error produces roughly what positioning error on the ground?

A. About 3 metres
B. About 30 metres
C. About 300 metres ✓
D. About 3 kilometres

Answer: C. About 300 metres

Distance is the speed of light multiplied by time, and light travels about 300 metres in a microsecond. This is why the satellites carry atomic clocks accurate to nanoseconds.

124. How do GPS satellites share the same frequencies without interfering?

A. Each transmits in a different time slot on a rotating schedule
B. Each is allocated a narrow sub-band within the L1 carrier
C. Each transmits only when it is above a set elevation angle
D. Each transmits a unique pseudo-random noise code, so the receiver separates them by code ✓

Answer: D. Each transmits a unique pseudo-random noise code, so the receiver separates them by code

This is code division multiple access. All the satellites occupy the same spectrum at once, and the receiver correlates against a known PRN code to pull out one satellite at a time.

125. Ephemeris and almanac data are both broadcast. What is the difference?

A. Ephemeris gives one satellite's precise orbit and is valid about four hours; the almanac gives approximate data for all satellites ✓
B. Ephemeris gives the clock correction and the almanac gives the orbit
C. Ephemeris is encrypted for military users and the almanac is open
D. Ephemeris is broadcast by GPS and the almanac by Galileo

Answer: A. Ephemeris gives one satellite's precise orbit and is valid about four hours; the almanac gives approximate data for all satellites

The almanac is coarse and covers the whole constellation, which is what lets a receiver work out which satellites to look for. The ephemeris is precise, applies to the transmitting satellite and expires quickly.

126. Which pair of orbital figures correctly separates GPS from Galileo?

A. GPS 23,222 km in 3 planes; Galileo about 20,200 km in 6 planes
B. GPS about 20,200 km in 6 planes; Galileo 23,222 km in 3 planes ✓
C. GPS about 20,200 km in 3 planes; Galileo 35,786 km in 6 planes
D. Both about 20,200 km, but GPS in 6 planes and Galileo in 12

Answer: B. GPS about 20,200 km in 6 planes; Galileo 23,222 km in 3 planes

Galileo flies higher and uses fewer, more populated planes, with a 24/3/1 Walker design. The altitude difference is also why the periods differ, about 12 hours against about 14 hours 5 minutes.

127. GPS L1 and Galileo E1 are both at 1575.42 MHz. What does that make possible?

A. The two systems can share one control segment
B. Galileo can correct GPS clock errors directly in orbit
C. A single receiver chip can track both constellations ✓
D. The two systems must coordinate their transmissions in time

Answer: C. A single receiver chip can track both constellations

Sharing the band and a compatible signal structure means one front end and one correlator bank serve both. That roughly doubles the satellites in view and lowers dilution of precision.

128. Which error source is described as amplifying all the others rather than adding its own delay?

A. Ionospheric delay
B. Multipath
C. Receiver noise and hardware imperfection
D. Geometric dilution of precision ✓

Answer: D. Geometric dilution of precision

Dilution of precision is about satellite geometry. Poorly spread satellites make the position solution ill-conditioned, so whatever ranging errors exist are magnified in the result.

129. Which error source is largest, and what is its quoted range?

A. Tropospheric delay, at about 2 to 25 metres ✓
B. Ionospheric delay, at about 1 to 2 metres
C. Multipath, at about 5 to 15 metres
D. Satellite clock and ephemeris error, at up to 1 metre

Answer: A. Tropospheric delay, at about 2 to 25 metres

The quoted figures are ionosphere 5 to 15 m, troposphere 2 to 25 m, multipath up to about 1 m, clock and ephemeris 1 to 2 m and receiver noise under 1 m. The troposphere has the widest top end, worst near the horizon.

130. Galileo's open service accuracy is quoted as better than GPS's traditional single-frequency figure. What is credited for the improvement?

A. A larger constellation of 32 operational satellites
B. Dual and multi-frequency signals plus OSNMA authentication, reaching about 20 cm with HAS ✓
C. A lower orbit that shortens the signal path
D. An encrypted military code available to civilian users

Answer: B. Dual and multi-frequency signals plus OSNMA authentication, reaching about 20 cm with HAS

The table quotes about 5 m for GPS single-frequency against under 1 m dual-frequency for Galileo. Multi-frequency operation cancels ionospheric delay, which is the dominant correctable error.

Post-quantum cryptography for non-terrestrial networks

Group 14 deck, slides 3 to 13

131. Which algorithm is the specific threat to RSA and ECC, and what does it do?

A. Grover's algorithm, which reverses a hash function directly
B. Shor's algorithm, which factors large primes efficiently on a quantum computer ✓
C. Shor's algorithm, which brute-forces a symmetric key in linear time
D. Grover's algorithm, which solves the discrete logarithm in constant time

Answer: B. Shor's algorithm, which factors large primes efficiently on a quantum computer

RSA and ECC rest on problems that are hard for classical computers. Shor's algorithm solves exactly those problems, which is why the public-key layer, not the symmetric layer, is the urgent one.

132. What is the harvest now, decrypt later threat?

A. A quantum computer decrypts traffic in real time as it is transmitted
B. Keys are stolen now and reused against future sessions
C. Traffic intercepted today is stored and decrypted once a quantum computer exists ✓

D. Encrypted data is corrupted now so it cannot be recovered later

Answer: C. Traffic intercepted today is stored and decrypted once a quantum computer exists

It makes the threat present rather than future. Data with a long secrecy lifetime is already at risk, because the attacker only needs to record it now and wait.

133. Which two algorithms are named as the post-quantum replacements?

A. RSA-4096 and ECC-521
B. AES-256 and SHA-256
C. IPsec and DTLS
D. CRYSTALS-Kyber and CRYSTALS-Dilithium ✓

Answer: D. CRYSTALS-Kyber and CRYSTALS-Dilithium

Kyber is the key encapsulation mechanism, standardised as NIST FIPS 203, and Dilithium is the signature scheme. The others in the list are either the vulnerable primitives or transport protocols.

134. Why is bandwidth overhead a bigger problem for PQC on a satellite link than on the ground?

A. PQC keys and signatures are far larger than RSA or ECC, and satellite capacity is limited and expensive ✓
B. Satellite links cannot carry packets above a fixed maximum size
C. PQC requires the key to be retransmitted on every packet
D. Satellite links have no error correction, so larger keys corrupt more often

Answer: A. PQC keys and signatures are far larger than RSA or ECC, and satellite capacity is limited and expensive

The size increase is the same everywhere. What differs is the cost of carrying it: on a satellite link every extra byte competes with revenue traffic on a scarce resource.

135. Why does propagation delay hurt PQC handshakes in particular?

A. The keys expire before they reach the far end
B. Multi-round-trip key exchanges multiply an already long round-trip time ✓
C. The satellite cannot buffer a handshake for longer than one pass
D. Delay causes the signature timestamp to fall outside its validity window

Answer: B. Multi-round-trip key exchanges multiply an already long round-trip time

A handshake that needs several exchanges pays the round-trip cost once per exchange. On a link where one round trip is already long, that multiplication dominates setup and re-keying time.

136. What makes onboard compute a constraint for PQC specifically?

A. Satellite processors cannot execute lattice arithmetic at all
B. Onboard memory is erased whenever the satellite passes through eclipse
C. Satellite payloads use radiation-hardened, lower-power processors on tight energy budgets ✓
D. Radiation-hardened processors cannot store private keys securely

Answer: C. Satellite payloads use radiation-hardened, lower-power processors on tight energy budgets

Radiation hardening trades performance for reliability, so a space processor is generations behind a ground server. PQC is more demanding than the ECC and RSA primitives it replaces.

137. Which resolution is proposed for migrating without abandoning existing security?

A. Disabling encryption on the feeder link
B. Moving all key exchange to the ground segment
C. Replacing AES with a longer symmetric key
D. Hybrid cryptographic handshakes ✓

Answer: D. Hybrid cryptographic handshakes

A hybrid handshake runs a classical and a post-quantum exchange together, so the session stays secure if either one holds. That is what IETF RFC 9370 provides for in IKEv2.

138. Subscriber identity privacy, primary authentication and air-interface ciphering are described as protecting which link?

A. The service link between the handset and the satellite ✓
B. The feeder link between the satellite and the ground station
C. The inter-satellite link within the constellation
D. The connection between the gateway and the 5G core

Answer: A. The service link between the handset and the satellite

Those three are the handset-facing mechanisms. The feeder link is protected instead by IPsec tunnels, DTLS for fast signalling and NDS/IP at the ground station.

139. Which mechanisms are named as protecting the path from the gateway to the core?

A. Subscriber identity privacy, primary authentication and air-interface ciphering
B. Service-based architecture security, GTP-U encryption and cross-network integrity ✓
C. IPsec tunnels, DTLS and NDS/IP border protection
D. Kyber key encapsulation, Dilithium signatures and AES-256

Answer: B. Service-based architecture security, GTP-U encryption and cross-network integrity

Once traffic is on the ground it moves through the core's own service-based architecture. SBI security, GTP-U tunnel encryption and integrity stamps on system messages guard that segment.

140. What does the conclusion identify as the reason PQC adoption is essential for space networks?

A. Satellite links are already broken by existing classical attacks
B. Post-quantum algorithms are faster than RSA on space processors
C. 5G NTNs run on standard classical cryptography today, which quantum computing threatens ✓
D. 3GPP has already mandated PQC in Release 17

Answer: C. 5G NTNs run on standard classical cryptography today, which quantum computing threatens

The argument is about exposure, not about current failure. The cryptography in use today is sound against classical attack and vulnerable to a quantum one, and satellites are hard to upgrade after launch.

Doppler shift estimation in 5G NR non-terrestrial networks

Group 15 deck, slides 3 to 14

141. For an S-band LEO satellite at 600 km, what is the quoted peak Doppler shift and where in the pass does the shift cross zero?

A. Up to about plus or minus 48 kHz, crossing zero at the horizon
B. Up to about plus or minus 5 kHz, crossing zero at the zenith
C. Up to about plus or minus 48 kHz, crossing zero at the zenith ✓
D. Up to about plus or minus 500 kHz, crossing zero at the horizon

Answer: C. Up to about plus or minus 48 kHz, crossing zero at the zenith

The shift is largest approaching the horizon, where the satellite is closing fastest along the line of sight, and passes through zero overhead, where the radial component vanishes.

142. Why does a transparent architecture suffer a doubled Doppler impact?

A. The signal is transmitted twice, once by the satellite and once by the gateway
B. The uplink and downlink Doppler shifts have the same sign and add
C. The bent-pipe repeater doubles the carrier frequency
D. Both the service link and the feeder link are Doppler-affected ✓

Answer: D. Both the service link and the feeder link are Doppler-affected

A regenerative payload demodulates on board, so Doppler is confined to the service link and the digital backhaul is clean. A transparent payload passes the signal through, so the two segments accumulate.

143. What does an uncompensated Doppler shift do to an OFDM signal?

A. It destroys subcarrier orthogonality and causes inter-carrier interference ✓
B. It shortens the cyclic prefix below the delay spread
C. It inverts the constellation mapping on every subcarrier
D. It causes the frame timing to drift out of the transmission window

Answer: A. It destroys subcarrier orthogonality and causes inter-carrier interference

OFDM works because each subcarrier peak sits on its neighbours' zero crossings. A frequency offset moves the peaks off those nulls, so each subcarrier leaks into the ones beside it.

144. On the downlink, one common signal is broadcast to all users in a beam. Why do they not all see the same Doppler shift?

A. Each user's receiver applies a different local oscillator correction

B. Each user has a different local elevation angle to the satellite ✓

C. The satellite transmits at a different frequency to each user

D. Users at the beam edge receive a delayed copy of the signal

Answer: B. Each user has a different local elevation angle to the satellite

Doppler depends on the radial component of velocity, which depends on where the user sits in the beam. The deck shows users at plus 40 kHz, 0 and minus 40 kHz within one footprint.

145. What system assumption does the 3GPP Release 17 baseline make?

A. Every NTN user device has a stable atomic reference oscillator

B. The satellite payload is always regenerative

C. Every NTN user device carries an active GNSS receiver ✓

D. The network broadcasts a single common Doppler value per beam

Answer: C. Every NTN user device carries an active GNSS receiver

The Release 17 workflow is ephemeris broadcast, UE self-positioning, Doppler calculation, uplink pre-compensation. Every step after the first depends on the device knowing its own position from GNSS.

146. Which of the following is given as a reason GNSS-assisted methods fail in practice?

A. GNSS satellites and NTN satellites use incompatible frequency bands

B. GNSS ephemeris cannot be broadcast over an NTN downlink

C. GNSS position fixes are too coarse to compute a Doppler shift

D. GNSS signals at about minus 130 dBm fall over in urban canyons, indoors and under foliage ✓

Answer: D. GNSS signals at about minus 130 dBm fall over in urban canyons, indoors and under foliage

The four failure reasons are environmental vulnerability, susceptibility to jamming and spoofing, low-cost IoT devices lacking GNSS chipsets, and oscillator error mixing with true Doppler.

147. Total Doppler is split as an integer subcarrier shift plus a fractional shift. What bounds the fractional part?

A. Its magnitude is less than 0.5 of the subcarrier spacing ✓

B. Its magnitude is less than one full subcarrier spacing

C. Its magnitude is less than 0.5 of the carrier frequency

D. It is bounded by the length of the cyclic prefix

Answer: A. Its magnitude is less than 0.5 of the subcarrier spacing

Beyond half a subcarrier the offset is better described as the next integer shift. Splitting the problem this way lets a coarse method find the integer and a fine method find the remainder.

148. How is the fractional Doppler recovered from the cyclic prefix?

A. The cyclic prefix carries a pilot tone at a known frequency

B. The cyclic prefix repeats the symbol's last samples, so a complex-conjugate product gives a phase angle ✓

C. The cyclic prefix is correlated against the PSS and SSS sequences

D. The cyclic prefix length is measured and compared with the nominal value

Answer: B. The cyclic prefix repeats the symbol's last samples, so a complex-conjugate product gives a phase angle

Because the prefix is an exact copy of the symbol tail, multiplying the signal by a delayed conjugate of itself yields a phase difference that is proportional to the frequency offset.

149. How is the integer Doppler shift recovered?

A. By reading the value from the ephemeris broadcast in SIB19

B. By measuring the phase rotation across the cyclic prefix

C. By frequency-domain cross-correlation of PSS and SSS over a sliding frequency search grid ✓

D. By comparing the arrival time of two consecutive frames

Answer: C. By frequency-domain cross-correlation of PSS and SSS over a sliding frequency search grid

The synchronisation signals are known sequences, so searching across candidate frequency offsets and taking the strongest correlation identifies which whole subcarrier the signal has moved by.

150. In the trade-off table, which method has no GNSS dependency and the lowest pilot overhead?

A. 3GPP Release 17 GNSS-based, which needs only minimal pilots

B. Multi-frequency pilots, which use DM-RS

C. AI / EKF tracking, which is fully GNSS-independent

D. CP / cross-correlation, which uses the standard cyclic prefix ✓

Answer: D. CP / cross-correlation, which uses the standard cyclic prefix

The cyclic prefix is already in every symbol, so exploiting it adds nothing to the transmission. Multi-frequency pilots are also GNSS-free but spend DM-RS resources, and EKF tracking may take an initial GNSS seed.

Network slicing in non-terrestrial networks

Group 16 deck, slides 3 to 10

151. What is network slicing?

A. Dividing the satellite footprint into smaller geographic cells

B. Splitting the frequency band into fixed sub-bands per operator

C. Separating the control plane from the user plane in the core

D. Running several independent virtual networks on one shared physical network, each tuned end to end ✓

Answer: D. Running several independent virtual networks on one shared physical network, each tuned end to end

The physical resources, that is spectrum, payload, transport and core compute, stay shared. What is divided is the logical treatment, so each slice behaves like a network of its own.

152. Which slice types match which service profiles?

A. SST 1 is eMBB with wide bandwidth, SST 2 is URLLC with priority scheduling, SST 3 is mMTC with small packets ✓

B. SST 1 is URLLC, SST 2 is mMTC, SST 3 is eMBB

C. SST 1 is mMTC, SST 2 is eMBB, SST 3 is URLLC

D. SST 1 is eMBB, SST 2 is mMTC, SST 3 is URLLC

Answer: A. SST 1 is eMBB with wide bandwidth, SST 2 is URLLC with priority scheduling, SST 3 is mMTC with small packets

eMBB is the wide, delay-tolerant pipe for video and broadband. URLLC needs short queues and guaranteed priority. mMTC is store-and-forward for huge numbers of tiny IoT packets.

153. Round-trip delay is quoted as 25.8 ms for a 600 km LEO and 541 ms for GEO. What does that break?

A. The eMBB slice cannot reach its throughput target

B. A 1 ms URLLC service level agreement simply cannot be honoured ✓

C. The mMTC slice cannot support enough simultaneous devices

D. The slice identifier cannot be carried in the packet header

Answer: B. A 1 ms URLLC service level agreement simply cannot be honoured

URLLC is defined by a latency bound of about 1 ms on a terrestrial cell. Physics puts the satellite round trip an order of magnitude or more above that before any processing is added.

154. Why is hard isolation between slices not available on a satellite payload?

A. 3GPP forbids physical isolation in non-terrestrial deployments
B. The slice identifier is stripped at the feeder link
C. Mass, power and thermal limits mean one payload and one spectrum pool are shared, so isolation has to be logical ✓
D. Satellites cannot run virtualisation software in orbit

Answer: C. Mass, power and thermal limits mean one payload and one spectrum pool are shared, so isolation has to be logical

On the ground a tenant can be given dedicated hardware. A satellite cannot carry duplicate payloads for each slice, so separation is enforced in software over shared resources.

155. In what sense does an NTN slice sit on a cell that will not stay still?

A. The user devices move between cells faster than the slice can be reconfigured
B. The slice identifier changes each time the satellite changes beam
C. The cell size shrinks as the satellite approaches the horizon
D. A LEO beam sweeps the ground and is visible for only 6 to 10 minutes, so the slice must be re-anchored to a new satellite ✓

Answer: D. A LEO beam sweeps the ground and is visible for only 6 to 10 minutes, so the slice must be re-anchored to a new satellite

In terrestrial 5G a slice is pinned to a fixed cell and a fixed fibre path. In NTN the radio cell moves out of view within minutes, so the anchoring has to be recomputed continuously.

156. What does it mean that the topology re-wires itself?

A. Feeder-link switchover and inter-satellite re-routing change the end-to-end path mid-session ✓
B. The constellation adds and removes satellites during operation
C. The user device switches between terrestrial and satellite access
D. The core network functions migrate between data centres

Answer: A. Feeder-link switchover and inter-satellite re-routing change the end-to-end path mid-session

Both the radio cell and the transport path are in motion. A slice's transport segment therefore keeps changing underneath it, which is not something terrestrial slicing has to handle.

157. In the proposed management chain, which function handles the beam and the scheduler?

A. CSMF, which captures customer intent
B. NSSMF (RAN) ✓
C. NSMF, which manages the end-to-end slice
D. NSSMF (CN/TN), which covers core and transport

Answer: B. NSSMF (RAN)

The chain runs CSMF for customer intent, NSMF for the end-to-end slice, then two subnet managers: one for the radio segment including beam and scheduler, and one for core and transport.

158. The orchestration is described as orbit-aware. What does that mean?

A. Slices are allocated according to the orbital altitude of each satellite
B. Each orbital plane is assigned its own dedicated slice
C. Slice policy is driven by ephemeris and handover timing rather than a static cell plan ✓
D. The orchestrator runs on board the satellite rather than on the ground

Answer: C. Slice policy is driven by ephemeris and handover timing rather than a static cell plan

Because the cell and the transport path move predictably, the orchestrator can plan against the constellation's known motion. The summary puts it as planning from ephemeris, not from a static cell plan.

159. Which cost is paired with the benefit of strict isolation between tenants?

A. Slice headers and per-slice buffering inflate packets on a long link
B. SLAs must be recomputed at every handover and feeder-link switchover
C. A regenerative gNB and UPF per slice generate heat the platform cannot shed
D. Capacity fenced off for an idle slice is wasted on a scarce, power-limited payload ✓

Answer: D. Capacity fenced off for an idle slice is wasted on a scarce, power-limited payload

Each benefit in the deck has its own cost. Isolation reserves capacity whether or not it is used, and on a payload with a hard power budget idle reserved capacity is pure loss.

160. The summary states that every gain has a price. How is that price characterised?

- **A. Isolation costs capacity and signalling overhead on a link that is already power-limited and delay-limited ✓**
- B. Slicing requires a larger constellation to deliver the same coverage
- C. Slicing forces all traffic through the ground core rather than on board
- D. Slicing prevents the use of regenerative payloads entirely

Answer: A. Isolation costs capacity and signalling overhead on a link that is already power-limited and delay-limited

The deck frames slicing in NTN as a budgeting exercise. The idea is the same as on the ground, but the physics is harder and every logical guarantee is drawn from a scarce physical budget.

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eRACH, a learned random access protocol for LEO networks

Group 17 deck, slides 3 to 8

161. What makes the topology of a LEO network non-stationary from the point of view of random access?

- **A. Satellites move at about 7.6 km/s, so the best satellite to connect to changes every few milliseconds ✓**
- B. The number of satellites in view changes as the constellation is expanded
- C. Ground terminals move between beams faster than the network can track
- D. The preamble set is re-randomised by the network at every access window

Answer: A. Satellites move at about 7.6 km/s, so the best satellite to connect to changes every few milliseconds

Standard RACH assumes a fixed tower and has no notion of which base station to associate with. In LEO that choice exists, it matters, and it changes continuously.

162. What is the quoted one-way propagation delay for LEO, and why does it matter for acknowledgement-based protocols?

- A. 1 to 5 ms, which is short enough that acknowledgement schemes work unchanged
- **B. 1 to 5 ms, which makes CSMA/CA and acknowledgement schemes slow and prone to wasted retransmissions ✓**
- C. 20 to 60 ms, which forces every protocol to abandon acknowledgements entirely
- D. under 1 ms, so the delay affects throughput but not protocol design

Answer: B. 1 to 5 ms, which makes CSMA/CA and acknowledgement schemes slow and prone to wasted retransmissions

Even LEO's relatively short delay is long compared with the time a terrestrial protocol expects to wait. Protocols built on listen-then-send or send-then-acknowledge degrade under it.

163. In eRACH, what does each ground terminal observe locally?

- A. The queue length and preamble choice of every neighbouring terminal
- B. A schedule broadcast by the network at the start of each access window
- **C. The expected satellite position from the known orbit, and whether its last attempt collided ✓**
- D. The aggregate collision rate measured across the whole beam

Answer: C. The expected satellite position from the known orbit, and whether its last attempt collided

The design point is that terminals do not talk to each other. Each has only what it can see for itself: predictable geometry, and the outcome of its own last attempt.

164. What decision does the actor-critic network make at each terminal?

- A. Which preamble index to select from the shared pool
- B. How much transmit power to apply to the preamble
- C. Whether to use the terrestrial network instead of the satellite
- **D. Whether to access now and which satellite to use, or to back off and wait ✓**

Answer: D. Whether to access now and which satellite to use, or to back off and wait

The action space combines timing and association. Choosing when to transmit and which satellite to aim at is precisely what standard RACH leaves unspecified in a moving constellation.

165. What does emergent mean in the name emergent RACH?

- **A. The coordinated access pattern arises from independent learning, with no coordination signalling ✓**
- B. The protocol emerges from the 3GPP standardisation process over several releases
- C. The satellite emerges into view and broadcasts the access schedule
- D. New preambles are generated on demand as devices join the network

Answer: A. The coordinated access pattern arises from independent learning, with no coordination signalling

Terminals never communicate with each other. Over many orbital cycles they nonetheless learn to spread their attempts apart, so the coordination is a product of learning rather than of messaging.

166. Which quantitative result is reported for eRACH against standard RACH?

- A. 31.2 percent higher throughput and about four times lower access delay
- **B. 54.6 percent higher throughput and about twice lower access delay ✓**
- C. 54.6 percent lower collision rate and about twice higher throughput
- D. Twice the throughput and 54.6 percent lower access delay

Answer: B. 54.6 percent higher throughput and about twice lower access delay

The summary cites Table II of the source paper. The gain is achieved with no inter-device coordination, which is the point being made.

167. What does the parameter rho control in eRACH?

- A. The learning rate of the actor-critic network
- B. The number of satellites a terminal may consider
- **C. The balance between maximising throughput and avoiding collisions ✓**
- D. The length of the backoff window in milliseconds

Answer: C. The balance between maximising throughput and avoiding collisions

Rho equals 0 tunes for throughput and rho equals 2 tunes for collision avoidance. That lets one protocol serve application classes with different tolerances.

168. Which challenge concerns the hardware rather than the algorithm?

- A. Sensitivity to satellite positioning accuracy
- B. A higher collision tolerance than standard RACH
- C. The absence of inter-device coordination
- **D. High computational complexity against size, weight and power constraints ✓**

Answer: D. High computational complexity against size, weight and power constraints

Running a neural policy on a terminal or a payload costs compute, and both are budget-limited. The other two are properties of how the learned policy behaves.

169. Why is sensitivity to satellite positioning accuracy a concern for eRACH?

- **A. One of the two local observations is the expected satellite position, so an error corrupts the policy input ✓**
- B. The terminal must report its own position to the satellite before access
- C. Positioning error changes the preamble that the terminal selects
- D. The satellite cannot compute the reward without an accurate position

Answer: A. One of the two local observations is the expected satellite position, so an error corrupts the policy input

The policy acts on what it observes. If the predicted geometry is wrong, the decision about when and where to transmit is made on a false picture. The deck notes the protocol is nonetheless robust to such errors.

170. What is the headline problem statement in the conclusion?

- A. Standard RACH cannot operate at all above a certain orbital velocity
- **B. Standard RACH ignores satellite mobility, which produces high collision rates and long access delays ✓**
- C. LEO constellations have too few preambles for the number of devices

D. Multi-agent learning is required by the 3GPP NTN specification

Answer: B. Standard RACH ignores satellite mobility, which produces high collision rates and long access delays

The deck frames the gap as a mismatch: a protocol designed for stationary towers is being used where the tower moves, and the resulting collisions and delays are the symptom.

RIS-enhanced NTN for coverage and capacity in 6G

Group 18 deck, slides 2 to 10

171. What is the coverage problem that a reconfigurable intelligent surface is meant to solve?

A. The satellite beam is too wide to serve individual users efficiently

B. A satellite needs a clear view of the user, and a building or a hill removes the link completely ✓

C. The satellite cannot generate enough beams for all the users in a cell

D. Atmospheric absorption removes the link at high frequencies

Answer: B. A satellite needs a clear view of the user, and a building or a hill removes the link completely

The deck shows a user in a building's shadow with no line of sight. The link budget is already tight, so a blocked path is not a degraded link, it is no link.

172. How does a RIS steer a reflection without any RF chain?

A. Each element amplifies the incident signal before re-radiating it

B. The surface physically rotates to face the intended receiver

C. Each element applies its own phase shift, so the reflected waves add up in one chosen direction ✓

D. The surface converts the signal to a different frequency before reflecting it

Answer: C. Each element applies its own phase shift, so the reflected waves add up in one chosen direction

This is passive beamforming. Controlling the relative phase across the aperture decides where the reflected wavefronts interfere constructively, and none of it requires a transmitter.

173. What distinguishes an active RIS from a passive RIS?

A. It uses discrete rather than continuous phase shifts

B. It responds differently at different frequencies

C. It is mounted on the satellite rather than on a building

D. Each element is connected to an amplifier and an RF chain, so amplitude as well as phase can be controlled ✓

Answer: D. Each element is connected to an amplifier and an RF chain, so amplitude as well as phase can be controlled

A passive RIS only adjusts phase, and sometimes amplitude, of what it reflects. An active RIS adds gain, which costs power and complexity but relieves the multiplied path loss.

174. Which RIS type uses discrete phase shifts such as 1-bit values of 0 or 180 degrees?

A. Digital RIS ✓

B. Tunable analog RIS

C. Hybrid RIS

D. Frequency-selective RIS

Answer: A. Digital RIS

Digital RIS quantises the phase into a small number of states, which lowers complexity and cost. Tunable analog RIS uses varactors to give continuous phase at higher hardware cost.

175. Which RIS type is suited to wideband and multi-band operation?

A. Passive RIS, because it introduces no frequency-dependent gain

B. Frequency-selective RIS, whose elements respond differently across frequency ✓

C. Hybrid RIS, because it mixes passive and active elements

D. Digital RIS, because 1-bit control is frequency independent

Answer: B. Frequency-selective RIS, whose elements respond differently across frequency

Giving elements a designed frequency response lets one surface manipulate several bands at once, which a single fixed phase profile cannot do.

176. Why must a RIS panel be large, and why must it be close to one end of the link?

A. The panel must be larger than the wavelength of the incident signal
B. A large panel is needed to dissipate the heat from the controller
C. The two hops multiply rather than add, so the combined path loss is severe ✓
D. Regulations require a minimum aperture for reflecting satellite signals

Answer: C. The two hops multiply rather than add, so the combined path loss is severe

In a relay the two hops are separate links. In a passive reflection the losses compound, so the only remedies are a very large aperture and placing the surface near the transmitter or the receiver.

177. The deck states that a passive surface cannot measure anything itself. What problem does that create?

A. The surface cannot be assigned an identity in the network
B. The surface cannot verify that its reflection reached the user
C. The surface cannot be updated once it has been installed
D. The channel has to be estimated by other means before the phases can be computed ✓

Answer: D. The channel has to be estimated by other means before the phases can be computed

Setting the phases requires knowing the channel from the transmitter to the surface and from the surface to the user. A device that only reflects contributes no measurements toward that.

178. What is the near-field effect listed among the limitations?

A. A large panel breaks the simple distance model that far-field design assumes ✓
B. The panel heats the surrounding structure at close range
C. The panel reflects into its own controller at short distances
D. The panel cannot be used indoors because of wall reflections

Answer: A. A large panel breaks the simple distance model that far-field design assumes

Standard beamforming assumes a plane wave across the aperture. When the aperture is large relative to the distance, the wavefront is curved and the plane-wave approximation fails.

179. How does a RIS add capacity as well as coverage?

A. It doubles the bandwidth available on the existing path
B. It provides a second path, so a second spatial stream becomes possible ✓
C. It allows the satellite to reuse the same frequency in adjacent beams
D. It compresses the data before reflecting it onward

Answer: B. It provides a second path, so a second spatial stream becomes possible

Coverage comes from building a path where none existed. Capacity comes from that path being distinct from the direct one, which gives the channel matrix another dimension to carry a stream on.

180. What is the stated standardisation status of RIS?

A. Specified in 3GPP Release 18 as part of 5G-Advanced
B. Standardised by the ITU for satellite use only
C. A candidate for 6G, not present in any released standard ✓
D. Withdrawn from standardisation after Release 17 study items

Answer: C. A candidate for 6G, not present in any released standard

The deck is explicit that RIS is still before standardisation. It is presented as a promising technique rather than a deployable specification.

Uplink time synchronization for NTN without GNSS

Group 21 deck, slides 3 to 11

181. What system configuration does the deck assume throughout?

A. 5G NR with a transparent bent-pipe payload, in GEO, with SC-FDMA on the uplink
B. LTE with a regenerative payload in MEO, with CP-OFDM on the uplink

C. 5G NR with a regenerative payload carrying an onboard gNodeB, in LEO, with CP-OFDM on the uplink ✓
D. 5G NR with a transparent payload in LEO, with OFDMA on the downlink only

Answer: C. 5G NR with a regenerative payload carrying an onboard gNodeB, in LEO, with CP-OFDM on the uplink

The assumption matters because a regenerative payload puts the base station in orbit, so the timing reference the device must align to is on the satellite itself.

182. In the Release 17 dual-loop architecture, what are the three timing advance components?

A. An open-loop part, a closed-loop part and a Doppler pre-compensation part
B. A service link part, a feeder link part and an inter-satellite link part
C. A coarse part, a fine part and a fractional part
D. A network-broadcast common part, a UE-computed part from GNSS and ephemeris, and a closed-loop adjustment ✓

Answer: D. A network-broadcast common part, a UE-computed part from GNSS and ephemeris, and a closed-loop adjustment

TA common is broadcast and covers what is shared across the beam. TA UE is the device's own geometry. TA adj is the residual correction signalled by the gNB after measurement.

183. Without GNSS, which part of the Release 17 scheme fails first?

A. The open-loop equation, because the device does not know its position relative to the satellite ✓
B. The closed loop, because the gNB can no longer measure arrival time
C. The common broadcast, because the network cannot compute it
D. The cyclic prefix, because it can no longer absorb the delay spread

Answer: A. The open-loop equation, because the device does not know its position relative to the satellite

The device is described as a blind UE. It cannot compute a propagation delay from geometry it cannot observe, so it has no basis for its initial timing advance.

184. Why is falling back to closed-loop signalling alone not sufficient?

A. The closed loop only corrects frequency, not timing
B. The first transmissions arrive with timing errors far larger than the cyclic prefix ✓
C. The gNB cannot send a timing advance command to an unsynchronised device
D. The closed loop requires a GNSS timestamp in every command

Answer: B. The first transmissions arrive with timing errors far larger than the cyclic prefix

A closed loop can only refine an estimate that is already close enough to be received. With no starting estimate the signal lands outside the reception window and there is nothing to measure.

185. What is the key insight behind GNSS-time-free drift compensation?

A. The satellite broadcasts its own clock drift, which the device mirrors
B. Clock drift in the device oscillator is proportional to the propagation delay
C. Satellite motion changes the propagation delay, so tracking downlink arrival-time drift reveals the timing advance ✓
D. Arrival-time drift can be removed entirely by lengthening the cyclic prefix

Answer: C. Satellite motion changes the propagation delay, so tracking downlink arrival-time drift reveals the timing advance

The device cannot measure absolute distance, but it can watch how the arrival time of successive downlink slots shifts. That drift is produced by the changing range, so it carries the information needed.

186. In the Extended Kalman Filter method, what does the state vector contain?

A. UE velocity, satellite velocity and the relative Doppler shift
B. Timing advance, frequency offset and oscillator drift rate
C. Beam index, elevation angle and slant range
D. UE latitude, satellite longitude and the UE-to-satellite distance ✓

Answer: D. UE latitude, satellite longitude and the UE-to-satellite distance

The filter estimates the geometry the device cannot observe directly, and the timing advance follows from the estimated distance rather than from a GNSS fix.

187. What is the stated strength of the EKF approach?

- **A. Robust performance with rapid convergence to accurate values ✓**
- B. Zero computational cost on the user device
- C. Complete independence from the downlink signal
- D. Guaranteed accuracy at the beam edge and the beam centre alike

Answer: A. Robust performance with rapid convergence to accurate values

The filter runs a predict-then-update cycle, so each new measurement refines the estimate. That is what lets it converge quickly from a poor starting point.

188. What is the trade-off in the network-provided common timing advance method?

- A. It is accurate for LEO but fails entirely for GEO beams
- **B. A user near the beam centre gets an accurate value, while a user near the beam edge can be significantly off ✓**
- C. It is accurate at first but drifts as the satellite moves
- D. It works only for devices that already have a GNSS fix

Answer: B. A user near the beam centre gets an accurate value, while a user near the beam edge can be significantly off

One broadcast value cannot describe every point in a footprint. It is correct where it was computed for, and its error grows with distance from that reference point.

189. In the timing advance feedback loop, what does the gNB actually measure?

- A. The Doppler shift of the arriving uplink carrier
- B. The received power of the uplink relative to a target
- **C. Whether the uplink arrived too early, too late, or inside the expected reception window ✓**
- D. The number of retransmissions the device has attempted

Answer: C. Whether the uplink arrived too early, too late, or inside the expected reception window

The expected reception window covers one uplink slot. The gNB classifies the arrival against that window and sends an increase, decrease or keep command accordingly, looping until alignment holds.

190. What is the overall conclusion of the deck?

- A. GNSS is unavoidable for NTN uplink synchronization
- B. Uplink synchronization is unnecessary with a regenerative payload
- C. GNSS can be replaced only by an onboard atomic clock in the device
- **D. Alternative methods allow uplink synchronization without GNSS ✓**

Answer: D. Alternative methods allow uplink synchronization without GNSS

The deck presents four methods, each avoiding the GNSS dependency in a different way, and concludes that the Release 17 assumption is a design choice rather than a physical necessity.

Deep reinforcement learning for space-air-ground 6G resource allocation

Group 22 deck, slides 2 to 14

191. What is a space-air-ground integrated network?

- A. A satellite constellation that also carries an Earth observation payload
- B. A terrestrial network whose backhaul is provided by satellite
- C. A network in which every user device can act as a relay
- **D. A layered network combining satellites, aerial platforms and terrestrial infrastructure under one system ✓**

Answer: D. A layered network combining satellites, aerial platforms and terrestrial infrastructure under one system

The deck shows space, aerial and ground operation centres feeding one 6G core. The point is that all three tiers are managed together rather than as separate networks.

192. What is resource allocation in a SAGIN, and why is it needed?

- **A. Distributing limited communication and computing resources among users and platforms so the network stays fast and fair ✓**
- B. Assigning each satellite a fixed share of the available spectrum at launch

C. Deciding which ground stations will serve which orbital planes
D. Allocating storage on board each satellite for buffered traffic

Answer: A. Distributing limited communication and computing resources among users and platforms so the network stays fast and fair

Resources are scarce and demand keeps moving. Sharing them intelligently is what keeps performance and fairness acceptable as the geometry and the traffic change.

193. Which three challenges are listed for SAGIN 6G NTN?

A. Orbital debris, launch cost and regulatory approval
B. Spectrum scarcity and interference, limited onboard power and energy, and uneven shifting user demand ✓
C. Handover failure, packet loss and jitter
D. Key distribution, authentication and privacy leakage

Answer: B. Spectrum scarcity and interference, limited onboard power and energy, and uneven shifting user demand

All three are resource problems. There is not enough spectrum, not enough power, and the demand that has to be served with them will not stay still.

194. In the general DRL model, what constitutes the state?

A. The reward accumulated over the previous episode
B. The set of actions the agent has not yet tried
C. Available power, bandwidth, SINR and channel state information ✓
D. The orbital elements of every satellite in the constellation

Answer: C. Available power, bandwidth, SINR and channel state information

The state is what the agent observes about the environment before acting. Here that is the current resource and channel picture that the allocation decision has to be made from.

195. What does the action consist of in this framework?

A. Handover execution and beam shaping only
B. Reward shaping and policy update
C. Ephemeris broadcast and timing advance signalling
D. Channel allocation, power control and access point selection ✓

Answer: D. Channel allocation, power control and access point selection

The action is the decision the policy controller emits. Those three levers are what actually change how resources are distributed in the network.

196. What is the reward built from?

A. Quality of service, quality of experience and spectrum efficiency ✓
B. Collision rate, access delay and energy per bit
C. Throughput, latency and orbital altitude
D. Coverage area, satellite count and handover rate

Answer: A. Quality of service, quality of experience and spectrum efficiency

The reward defines what good allocation means. Combining service quality, experienced quality and efficient spectrum use captures both the user's view and the operator's view.

197. Which algorithm is described as choosing from a list of fixed options, such as which channel to use?

A. Deep Deterministic Policy Gradient
B. Deep Q-Network ✓
C. Proximal Policy Optimization
D. Soft Actor-Critic

Answer: B. Deep Q-Network

DQN works over a discrete action set. Continuous quantities such as an exact power level need DDPG or a similar method that outputs a real value rather than a choice.

198. Which algorithm is described as fine-tuning exact values such as a precise power level?

A. Deep Q-Network

B. Proximal Policy Optimization
C. Deep Deterministic Policy Gradient ✓
D. Twin Delayed DDPG

Answer: C. Deep Deterministic Policy Gradient

DDPG is the continuous-control counterpart to DQN. PPO is described as learning a stable strategy step by step, and SAC or TD3 as exploring boldly while converging reliably.

199. In the HAPS tier, which three allocation tasks are named?

A. Beam hopping, spectrum provisioning and transmit power allocation
B. Drone 3D location assignment, channel estimation and anti-interference management
C. Offloading, terrestrial base station selection and cloud routing
D. Position allocation, dynamic spectrum and power distribution, and edge server compute provisioning ✓

Answer: D. Position allocation, dynamic spectrum and power distribution, and edge server compute provisioning

Each tier has its own task list. Beam hopping and frequency assignment belong to the space tier, and 3D location assignment belongs to the UAV tier.

200. Which three challenges are listed for DRL-based SAGIN resource allocation?

A. Scalability, non-stationarity from high mobility, and real-time deadlines against slow learning ✓
B. Scarce training data, the sim-to-real gap and certification
C. Self-interference, synchronisation and onboard compute
D. Bandwidth overhead, handshake latency and radiation-hardened processors

Answer: A. Scalability, non-stationarity from high mobility, and real-time deadlines against slow learning

The environment is large, it will not hold still long enough for a policy to settle, and decisions are needed faster than learning naturally converges. Those three are what make the problem hard.

HARQ mechanisms and limitations in NTN

Group 23 deck, slides 2 to 14

201. How does HARQ differ from ARQ and from forward error correction alone?

A. It combines forward error correction with retransmission and soft combining of the failed copy ✓
B. It retransmits automatically without waiting for an acknowledgement
C. It corrects errors without any retransmission at all
D. It applies error correction only after a retransmission has failed

Answer: A. It combines forward error correction with retransmission and soft combining of the failed copy

ARQ waits for an acknowledgement and resends. FEC adds redundancy and decodes without resending. HARQ does both, and crucially keeps the failed copy so the two receptions can be combined.

202. In chase combining, what does the transmitter send on the retransmission?

A. A different set of parity bits from the same encoder
B. The same coded packet again, which the receiver combines with the stored soft bits ✓
C. Only the systematic bits, without any parity
D. A shorter packet containing just the bits that failed

Answer: B. The same coded packet again, which the receiver combines with the stored soft bits

The receiver keeps the soft bits from the first attempt rather than discarding them. Adding a second identical copy raises the effective signal-to-noise ratio enough to decode.

203. What distinguishes incremental redundancy from chase combining?

A. The retransmission is sent at a higher power level
B. The receiver discards the first copy before decoding the second
C. The retransmission carries different parity bits, so previously punctured bits are now sent ✓

D. The retransmission uses a different modulation scheme

Answer: C. The retransmission carries different parity bits, so previously punctured bits are now sent

The first transmission punctures some parity bits to raise the code rate. The retransmission supplies those bits, so the combined packet is protected by a lower-rate, stronger code.

204. Why does long round-trip time stall the HARQ pipeline in NTN?

A. The soft buffer overflows before the acknowledgement arrives
B. The acknowledgement expires and is discarded by the receiver
C. The receiver cannot decode a packet that arrives after a long delay
D. Every process stays occupied waiting for its acknowledgement, so no free process remains for new data ✓

Answer: D. Every process stays occupied waiting for its acknowledgement, so no free process remains for new data

A HARQ process is held from transmission until its acknowledgement returns. On a short terrestrial link they are released quickly. Over a 30 ms to 500 ms round trip they all fill up and transmission stops.

205. How many HARQ processes does 5G use by default, and what does the NTN adaptation raise that to?

A. 16 by default, raised to up to 32 for NTN ✓
B. 8 by default, raised to up to 16 for NTN
C. 16 by default, raised to up to 64 for NTN
D. 32 by default, raised to up to 64 for NTN

Answer: A. 16 by default, raised to up to 32 for NTN

More parallel processes keep the pipe full while earlier ones wait for acknowledgements. The cost is proportionally more soft buffer memory in the receiver.

206. What is feedback-less HARQ operation, and what does it rely on instead?

A. The device sends only NACKs, never ACKs, to halve the feedback load
B. The device sends no ACK or NACK, and the link relies on strong forward error correction or blind retransmission ✓
C. Feedback is sent to the gateway rather than to the satellite
D. Feedback is delayed until the end of the satellite pass

Answer: B. The device sends no ACK or NACK, and the link relies on strong forward error correction or blind retransmission

Removing the acknowledgement removes the waiting, which is what caused the stall. The redundancy has to be added in advance instead, because there is no longer any signal telling the sender what failed.

207. In HARQ-less operation, where is error recovery performed?

A. At the physical layer, using a stronger modulation scheme
B. At the application layer, by requesting the file again
C. At the RLC layer or above, which is slower but avoids stalling MAC and PHY ✓
D. On the satellite, which retransmits from its own buffer

Answer: C. At the RLC layer or above, which is slower but avoids stalling MAC and PHY

The MAC and PHY stop waiting for HARQ feedback entirely. Recovery still happens, but a layer up, which trades slower error recovery for a pipeline that keeps moving.

208. Which trade-off pair is stated correctly?

A. Longer timers give faster failure detection and use less memory
B. Disabling feedback gives faster error recovery and fewer false alarms
C. Onboard processing gives the slowest loop at the lowest satellite cost
D. More HARQ processes give higher throughput and require more memory ✓

Answer: D. More HARQ processes give higher throughput and require more memory

The four trade-offs are: more processes buy throughput with memory; longer timers avoid false alarms but slow failure detection; disabling feedback keeps flow constant but delays recovery; onboard processing gives the fastest loop at the highest satellite cost.

209. In a transparent payload, where does the gNB sit relative to the HARQ loop?

A. On the ground at the gateway, so the HARQ loop crosses both the service and feeder links ✓
B. On the satellite, so the HARQ loop covers only the service link
C. In the 5G core, so the loop crosses the N6 interface
D. Split between the satellite and the gateway, one process each

Answer: A. On the ground at the gateway, so the HARQ loop crosses both the service and feeder links

A transparent satellite is a repeater. The base station is at the gateway, so an acknowledgement travels user to satellite to gateway and back, which is the longest possible loop.

210. What does a regenerative payload change about the HARQ loop?

A. The loop is removed entirely because errors are corrected on board
B. The gNB is on the satellite, so the loop closes over the service link alone ✓
C. The loop is extended to include the inter-satellite link
D. The loop moves to the RLC layer automatically

Answer: B. The gNB is on the satellite, so the loop closes over the service link alone

Putting the base station in orbit halves the geometry the acknowledgement has to cross. That is the fastest loop, and the deck notes it is also the highest satellite cost.

Network digital twinning for 3D satellite constellation optimization

Group 24 deck, slides 3 to 9

211. What is a network digital twin?

A. A backup constellation held in reserve for failover
B. A virtual replica of the physical network that is continuously updated with real-world data ✓
C. A simulation model built once during the design phase
D. A duplicate of the network's control software running on the ground

Answer: B. A virtual replica of the physical network that is continuously updated with real-world data

The defining property is the live link back to reality. A model that is not continuously updated is a simulation; the continuous update is what makes it a twin.

212. Which of these is listed as a challenge that makes a 6G NTN constellation hard to manage?

A. The gradual decay of orbits over the mission lifetime
B. The limited number of ground stations licensed worldwide
C. Rapid topology changes, as links between satellites and ground stations are created and broken ✓
D. The fixed bandwidth allocation agreed at constellation design

Answer: C. Rapid topology changes, as links between satellites and ground stations are created and broken

The six challenges are constant mobility, rapid topology change, long propagation delay, limited onboard resources, frequent handovers and high deployment and maintenance cost.

213. What are the stated consequences of those challenges?

A. Orbital debris, collision risk and end-of-life disposal cost
B. Spectrum interference, cross-border regulation and licensing delay
C. Key compromise, replay attacks and identity exposure
D. Increased latency, routing instability, resource allocation difficulty, reduced QoS, slow fault detection and higher operational cost ✓

Answer: D. Increased latency, routing instability, resource allocation difficulty, reduced QoS, slow fault detection and higher operational cost

The consequence row follows directly from the challenge row: a network that changes every second is hard to route through, hard to allocate for and hard to diagnose.

214. In the hierarchical digital twin architecture, what does the network control centre hold?

A. A global controller and a central digital twin covering the whole network ✓

B. An edge digital twin for each ground station
C. The physical satellites and their telemetry links
D. A local controller managing beam and radio resource allocation

Answer: A. A global controller and a central digital twin covering the whole network

The hierarchy runs from edge twins at the ground stations, through local controllers, up to a central twin and a global controller at the network control centre.

215. Which functions belong to the local controller at a ground station?

A. Network verification, slicing and global optimization
B. Beam allocation, radio resource allocation, data processing and fault diagnosis ✓
C. Traffic engineering and global network modelling
D. Orbit determination and ephemeris broadcast

Answer: B. Beam allocation, radio resource allocation, data processing and fault diagnosis

The split is by scope. Local controllers handle what happens at one station, while the global controller handles network verification, slicing, traffic engineering and whole-network optimisation.

216. What are the five steps by which digital twinning optimizes a constellation?

A. Observe, decide, act, learn, repeat
B. Sense, optimize, act, learn, verify
C. Collect real-time data, synchronize the twin, simulate future conditions, evaluate options, deploy the best decision ✓
D. Predict, measure, correct, verify, log

Answer: C. Collect real-time data, synchronize the twin, simulate future conditions, evaluate options, deploy the best decision

The distinguishing step is simulating multiple futures before committing to one. That is what a twin adds over a plain control loop: decisions are tested in the model rather than on the live network.

217. Which benefits are attributed to the digital twin approach?

A. Higher orbital altitude, wider coverage and longer satellite life
B. Stronger encryption, better authentication and privacy protection
C. Larger constellations, cheaper launches and faster deployment
D. Lower latency, better routing, faster fault detection and reduced operational cost ✓

Answer: D. Lower latency, better routing, faster fault detection and reduced operational cost

All four follow from being able to test a decision before applying it: better routes are found, faults are noticed sooner, and fewer costly interventions are needed on the real network.

218. Why is DT migration complexity listed as a limitation?

A. Digital twins have to migrate as the satellites they model move ✓
B. Twins must be re-certified whenever the network software is updated
C. Twins cannot be transferred between different vendors' platforms
D. Twins lose their history whenever they are moved

Answer: A. Digital twins have to migrate as the satellites they model move

A twin is anchored to the physical thing it mirrors. When that thing is in orbit and the serving infrastructure changes, the twin has to follow it, which adds machinery the ground case does not need.

219. Why is synchronization overhead a real cost rather than an implementation detail?

A. Synchronization requires an atomic clock on every satellite
B. Keeping the physical and digital networks in step consumes bandwidth ✓
C. The twin cannot run while synchronization is in progress
D. Synchronization errors accumulate and cannot be corrected

Answer: B. Keeping the physical and digital networks in step consumes bandwidth

The twin is only useful while it matches reality, and matching reality means a continuous stream of telemetry. On a link where capacity is scarce, that stream competes with user traffic.

220. What is the security concern raised about digital twins?

A. The twin stores subscriber data that could be exposed
B. The twin's synchronization traffic cannot be encrypted

C. A compromised twin could be exploited to disrupt network operations ✓
D. The twin can be used to locate satellites for physical attack

Answer: C. A compromised twin could be exploited to disrupt network operations

The twin is not a passive mirror. Decisions computed in it are deployed to the physical network, so an attacker who controls the twin gains influence over the real constellation.

AI-driven predictive handover for high-mobility LEO networks

Unnumbered deck, AI-Driven Predictive Handover Management, slides 4 to 16

221. How does the handover situation in a LEO network differ from a terrestrial one?

A. On LEO the user must select the satellite manually before each session
B. On LEO handovers happen only when the user physically moves
C. On a normal mobile network you move past towers; on LEO the tower itself moves across the sky ✓
D. On LEO the handover is performed by the satellite without informing the device

Answer: C. On a normal mobile network you move past towers; on LEO the tower itself moves across the sky

This is the root of every other problem in the deck. A stationary user still gets handed over every few minutes, because the coverage is what is moving.

222. What is a handover storm?

A. A single device switching repeatedly between two satellites
B. A burst of signalling caused by a satellite failure
C. Handovers triggered by atmospheric interference during a storm
D. Many devices sharing one satellite all need to switch at nearly the same time ✓

Answer: D. Many devices sharing one satellite all need to switch at nearly the same time

The deck likens it to every shopper reaching the same checkout at once. Because they all share one satellite, they all lose it together, and the signalling arrives in one spike.

223. What is the ping-pong effect, and what causes it?

A. A device flips repeatedly between two satellites because the system is not confident in its decision ✓
B. A device alternates between the uplink and downlink of the same satellite
C. An acknowledgement bounces between the satellite and the gateway
D. A beam sweeps back and forth across the same ground cell

Answer: A. A device flips repeatedly between two satellites because the system is not confident in its decision

Each flip costs signalling and interrupts the connection, so the effect wastes effort and degrades service without ever improving the link.

224. Why is a reactive handover riskier in a satellite network than on the ground?

A. Satellite receivers cannot measure signal strength accurately
B. By the time the weak-signal warning reaches the network and a decision returns, the satellite has already moved a meaningful distance ✓
C. Ground networks do not use signal-strength triggers at all
D. The satellite cannot store the measurement report while in motion

Answer: B. By the time the weak-signal warning reaches the network and a decision returns, the satellite has already moved a meaningful distance

The deck contrasts a fixed tower, where only the radio signal is unpredictable, with a satellite, where the geometry itself changes during the decision. The decision arrives based on outdated information.

225. What is the key insight that makes a proactive approach possible?

A. Users follow predictable daily movement patterns
B. Radio interference in space is lower and therefore more stable
C. A satellite's flight path follows fixed, well-known laws of orbital motion, so it can be predicted ✓

D. Satellites broadcast their planned handover schedule in advance

Answer: C. A satellite's flight path follows fixed, well-known laws of orbital motion, so it can be predicted

The deck notes the same physics is used to predict eclipses and plan rocket launches. Rise, culmination and set times can all be computed ahead of the event.

226. What does conditional handover give the device, and what is its stated limitation?

A. A predicted handover time, but no backup target

B. A fully learned policy, but no fallback if prediction fails

C. A guaranteed target satellite, but only for stationary users

D. Backup satellites and switching conditions arranged in advance, but the conditions are relatively simple and fixed ✓

Answer: D. Backup satellites and switching conditions arranged in advance, but the conditions are relatively simple and fixed

Conditional handover removes the last-minute negotiation, which is a real improvement. The deck positions AI as filling the remaining gap by making those trigger conditions adaptive rather than fixed.

227. Layer 1 forecasts trajectory and channel quality. Which two approaches are named?

A. Forecasting physical quantities using propagation models such as SGP4, or forecasting the RSRP time series with CNN plus LSTM models ✓

B. Forecasting user demand with a Markov chain, or forecasting load with a Kalman filter

C. Forecasting orbital decay with numerical integration, or forecasting weather with a neural network

D. Forecasting beam occupancy with a scheduler, or forecasting handover counts with regression

Answer: A. Forecasting physical quantities using propagation models such as SGP4, or forecasting the RSRP time series with CNN plus LSTM models

The first predicts elevation angle, slant range, Doppler and path loss from known orbital mechanics. The second skips the physics and predicts the measured signal series directly.

228. In Layer 2 the handover is formulated as a Markov Decision Process. What is the action?

A. Deciding whether to send a measurement report

B. Choosing a target satellite, or staying with the current one ✓

C. Setting the transmit power for the next uplink slot

D. Selecting which beam of the serving satellite to use

Answer: B. Choosing a target satellite, or staying with the current one

State is the current and predicted link and geometry. Action is the association decision. Reward balances link quality against handover cost, penalising frequent handovers and failures.

229. How does DHO differ from the legacy protocol in the comparison shown?

A. It adds a second measurement report to improve confidence

B. It moves the handover decision from the network to the device

C. It removes the measurement report step entirely and predicts the handover action directly ✓

D. It replaces the reward function with a fixed threshold

Answer: C. It removes the measurement report step entirely and predicts the handover action directly

The legacy sequence is measure, report, decide, hand over. DHO observes, decides and hands over, which is where the lower access delay and lower collision rate come from.

230. Layer 3 is compared to an air-traffic controller. What does that analogy convey?

A. It gives priority to aviation users over other traffic

B. It routes each device along a fixed corridor through the constellation

C. It hands control of the decision to a human operator on the ground

D. It optimises the handover pattern across all users and satellites at once, rather than one connection in isolation ✓

Answer: D. It optimises the handover pattern across all users and satellites at once, rather than one connection in isolation

A controller sequences many aircraft onto runways so the whole system flows. The graph-based scheduler does the same for handovers, balancing load and avoiding congestion network-wide.

AI-assisted trajectory optimization of UAV and HAPS platforms

Unnumbered deck, AI-Assisted Trajectory Optimization, slides 3 to 11

231. What problem does AI-assisted trajectory optimization address in a 6G NTN?

A. Satellites cannot be launched into the orbits that 6G requires
B. Terrestrial base stations cannot support 6G modulation schemes
C. UAVs cannot legally operate in the same airspace as aircraft
D. Static deployment produces poor coverage, high latency and network congestion in areas that are hard to serve ✓

Answer: D. Static deployment produces poor coverage, high latency and network congestion in areas that are hard to serve

The platforms can move, but if they follow fixed paths they behave like fixed towers. Letting AI place and move them is what turns mobility into an advantage.

232. Which of these is listed as a 3D coverage challenge for current UAV and HAPS deployment?

A. Static trajectories that do not adapt to where users actually are ✓
B. The inability of UAVs to operate above 20 km altitude
C. The lack of a standardised air interface for aerial platforms
D. Excessive spectrum allocated to aerial platforms

Answer: A. Static trajectories that do not adapt to where users actually are

The six listed challenges are static trajectories, uneven user distribution, coverage holes, high interference, energy limitations and dynamic user mobility.

233. In the deck's terms, how do artificial intelligence and machine learning relate?

A. Artificial intelligence is a branch of machine learning focused on autonomy
B. Machine learning is a branch of AI in which machines learn from data instead of being explicitly programmed ✓
C. They are two names for the same set of techniques
D. Machine learning applies only to supervised problems, AI to unsupervised ones

Answer: B. Machine learning is a branch of AI in which machines learn from data instead of being explicitly programmed

AI is the broad goal of machines performing human-like tasks. Machine learning is the specific approach of improving performance from data rather than from hand-written rules.

234. Why is deep reinforcement learning chosen for this problem?

A. It requires no training data of any kind
B. It produces a fixed optimal trajectory that never needs updating
C. It learns by interacting with the environment, adapts to changing user locations and optimizes long-term reward ✓
D. It runs entirely on the ground, so the platform needs no compute

Answer: C. It learns by interacting with the environment, adapts to changing user locations and optimizes long-term reward

Trajectory planning is sequential: where you go now changes what is available later. Optimising long-term reward rather than the immediate step is exactly what reinforcement learning does.

235. Which of these does the platform collect as part of its state?

A. Preamble collisions, backoff windows and barring probabilities
B. Ephemeris, timing advance and Doppler shift
C. Slice identifiers, QoS flows and session contexts
D. User locations, signal strength, battery level, traffic demand, obstacles, weather and backhaul link quality ✓

Answer: D. User locations, signal strength, battery level, traffic demand, obstacles, weather and backhaul link quality

The state has to cover everything that should influence where the platform flies next, which is why it spans radio conditions, platform health, demand and the physical environment.

236. What does the reward function balance?

A. Maximising coverage while minimising energy, delay and interference ✓
B. Maximising altitude while minimising flight time
C. Maximising throughput while minimising the number of platforms
D. Maximising battery life while minimising the number of users served

Answer: A. Maximising coverage while minimising energy, delay and interference

Coverage alone would send the platform to an unsustainable position. The three penalties keep the solution within what the platform can physically sustain and what the network can tolerate.

237. Which outcomes earn a positive reward?

A. Higher altitude, wider beams and longer flight duration
B. More users covered, higher throughput, lower latency, lower interference and lower energy use ✓
C. Fewer handovers, fewer beams and lower bandwidth use
D. More platforms deployed and larger coverage radius

Answer: B. More users covered, higher throughput, lower latency, lower interference and lower energy use

The positive terms are the service outcomes the network wants. The negative terms are coverage holes, signal blockage, high interference, excessive battery use and weak backhaul.

238. In the layered architecture, what role do HAPS platforms play?

A. Low-altitude platforms that provide on-demand coverage of hotspots
B. Orbital platforms that provide wide-area broadcast services
C. Quasi-stationary platforms at about 20 km that act as aerial base stations and provide backhaul for UAVs ✓
D. Ground platforms that aggregate traffic from the terrestrial network

Answer: C. Quasi-stationary platforms at about 20 km that act as aerial base stations and provide backhaul for UAVs

The architecture stacks satellites for wide-area coverage, HAPS at about 20 km covering hundreds of kilometres, and low-altitude UAVs providing flexible on-demand capacity below.

239. Which is listed among the challenges of the approach?

A. The absence of any suitable learning algorithm
B. The inability to measure signal strength from an aerial platform
C. The lack of demand for coverage in remote areas
D. Airspace regulation ✓

Answer: D. Airspace regulation

The five challenges are battery limitations, AI computational complexity, airspace regulations, security and weather effects. Several are non-technical constraints on an otherwise workable technique.

240. What does the conclusion claim reinforcement learning enables?

A. Autonomous, adaptive decision-making that improves coverage, energy efficiency and quality of service ✓
B. The removal of all terrestrial infrastructure from the network
C. A single fixed trajectory that is optimal for every deployment
D. Coverage guarantees that do not depend on battery capacity

Answer: A. Autonomous, adaptive decision-making that improves coverage, energy efficiency and quality of service

The claim is about how the decision is made, not about removing constraints. The platform decides for itself and keeps improving, and battery and regulation remain real limits.